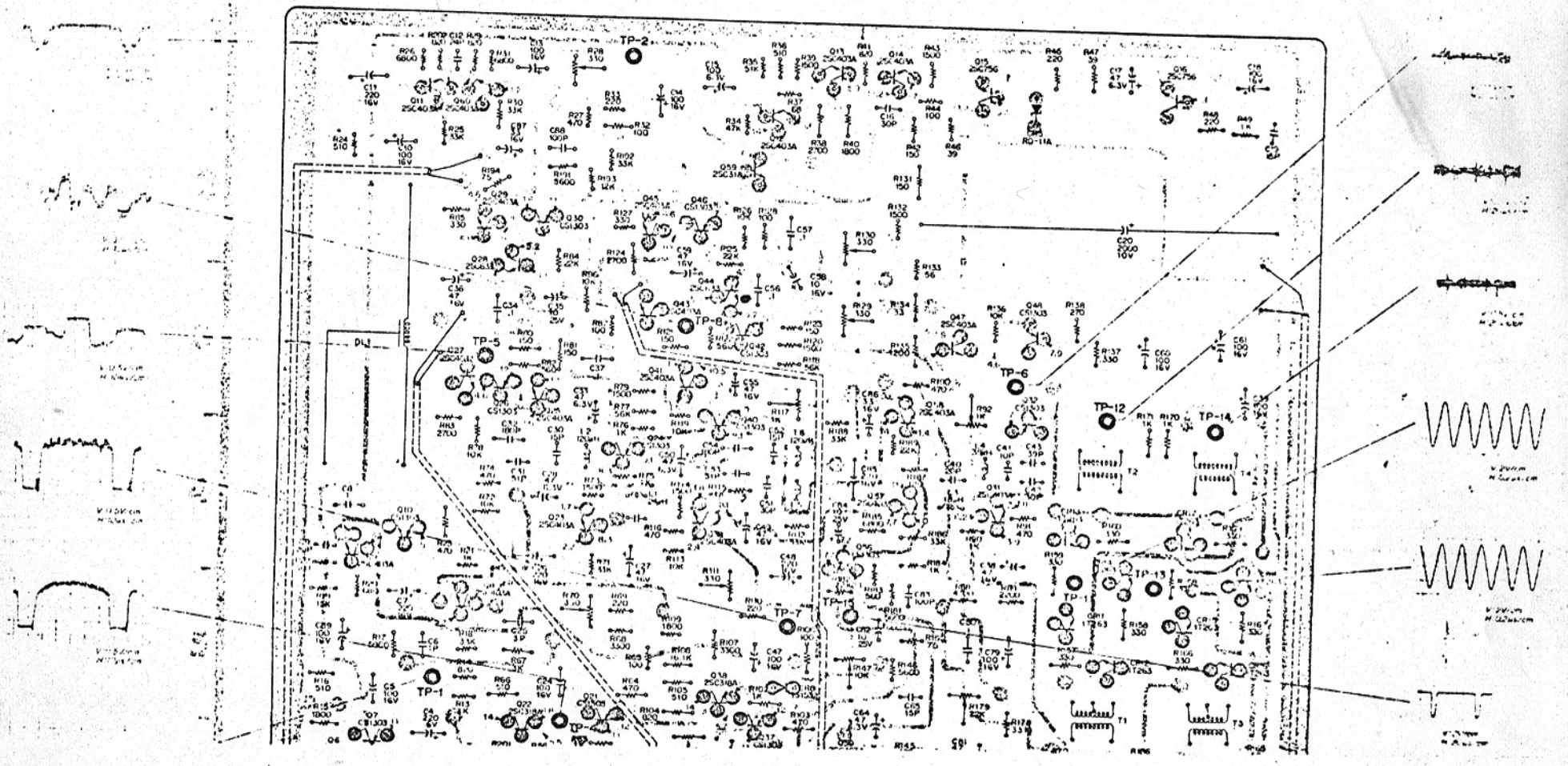
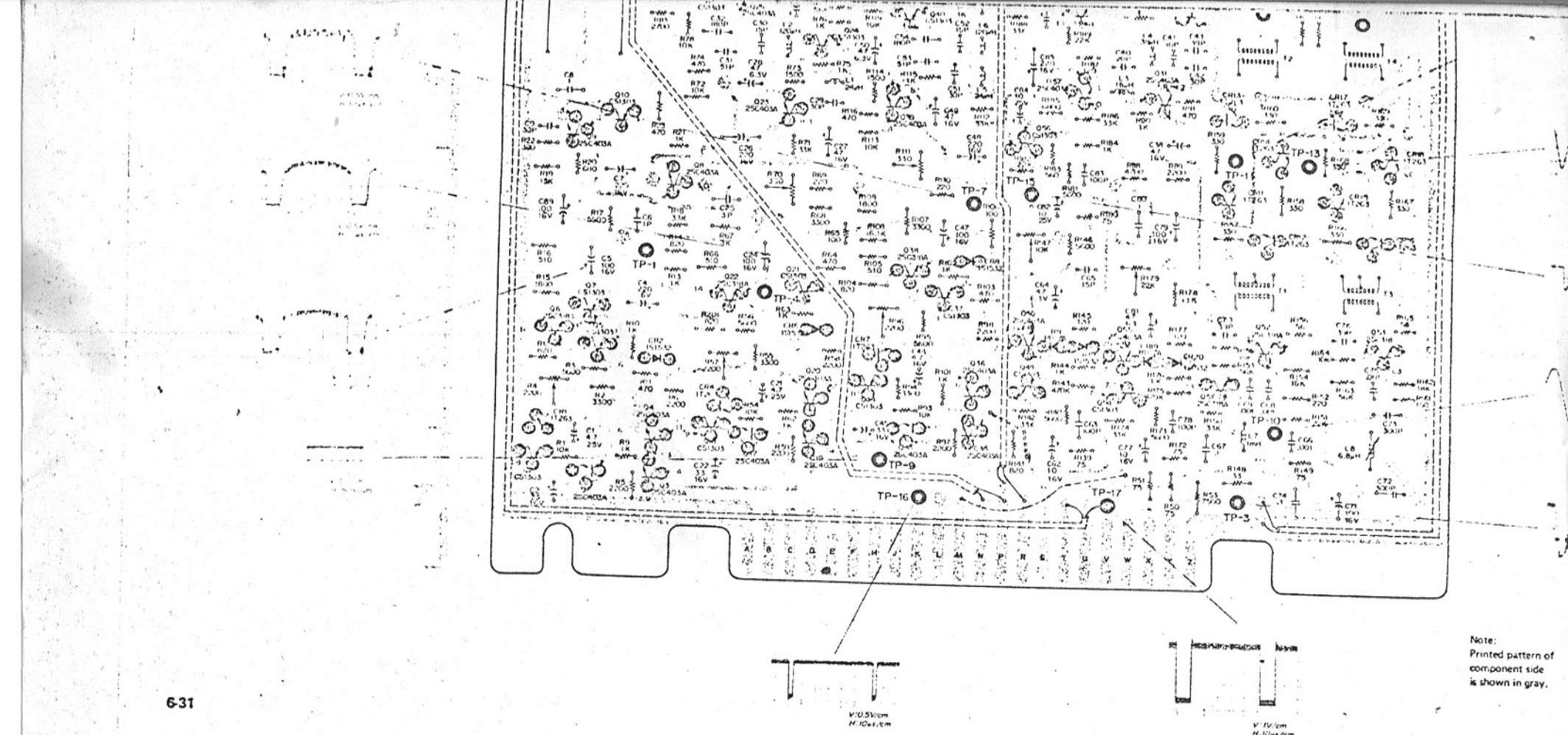
. ENCODER

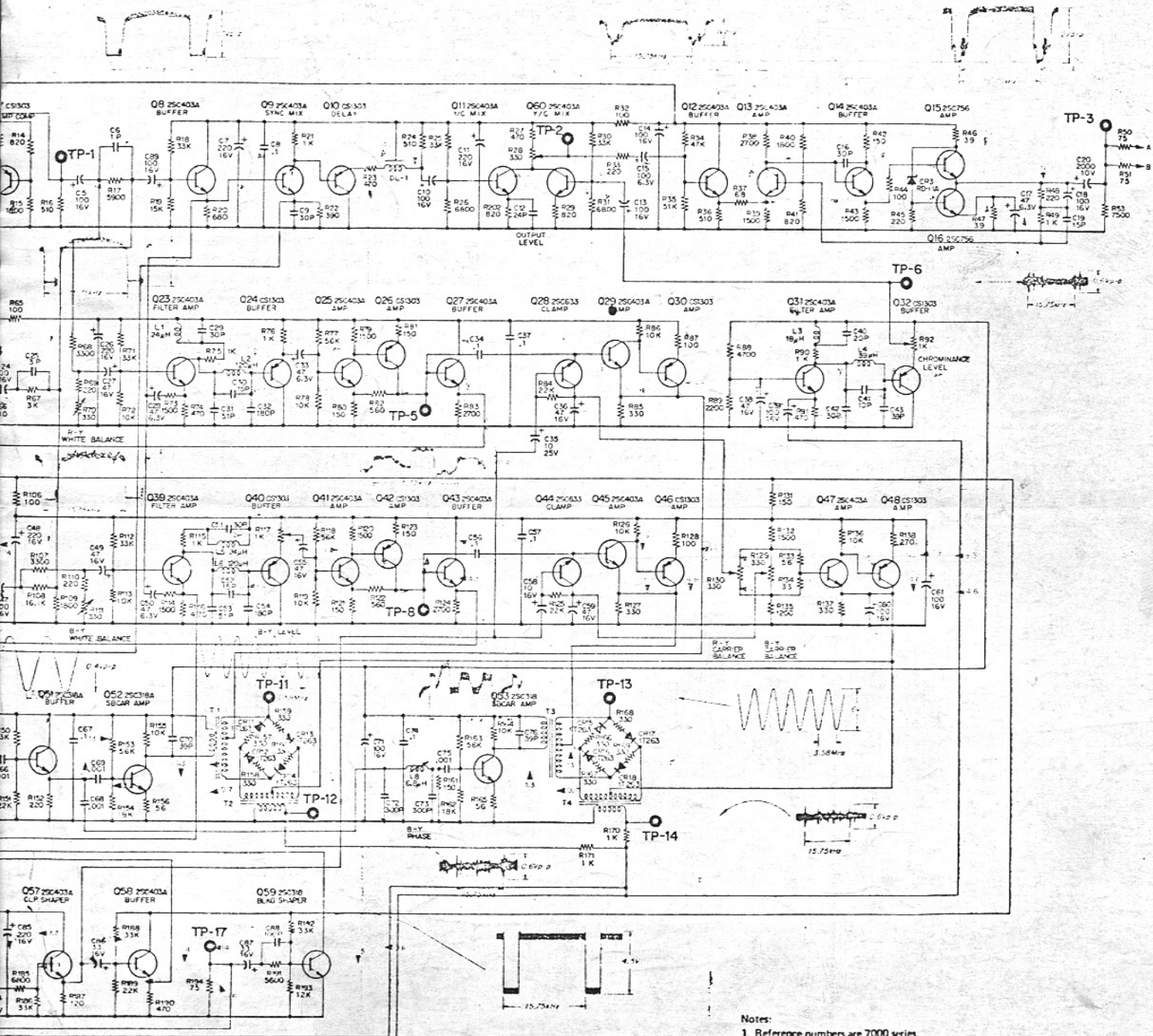
—conducter side—





Note:
Printed pattern of component side is shown in gray.

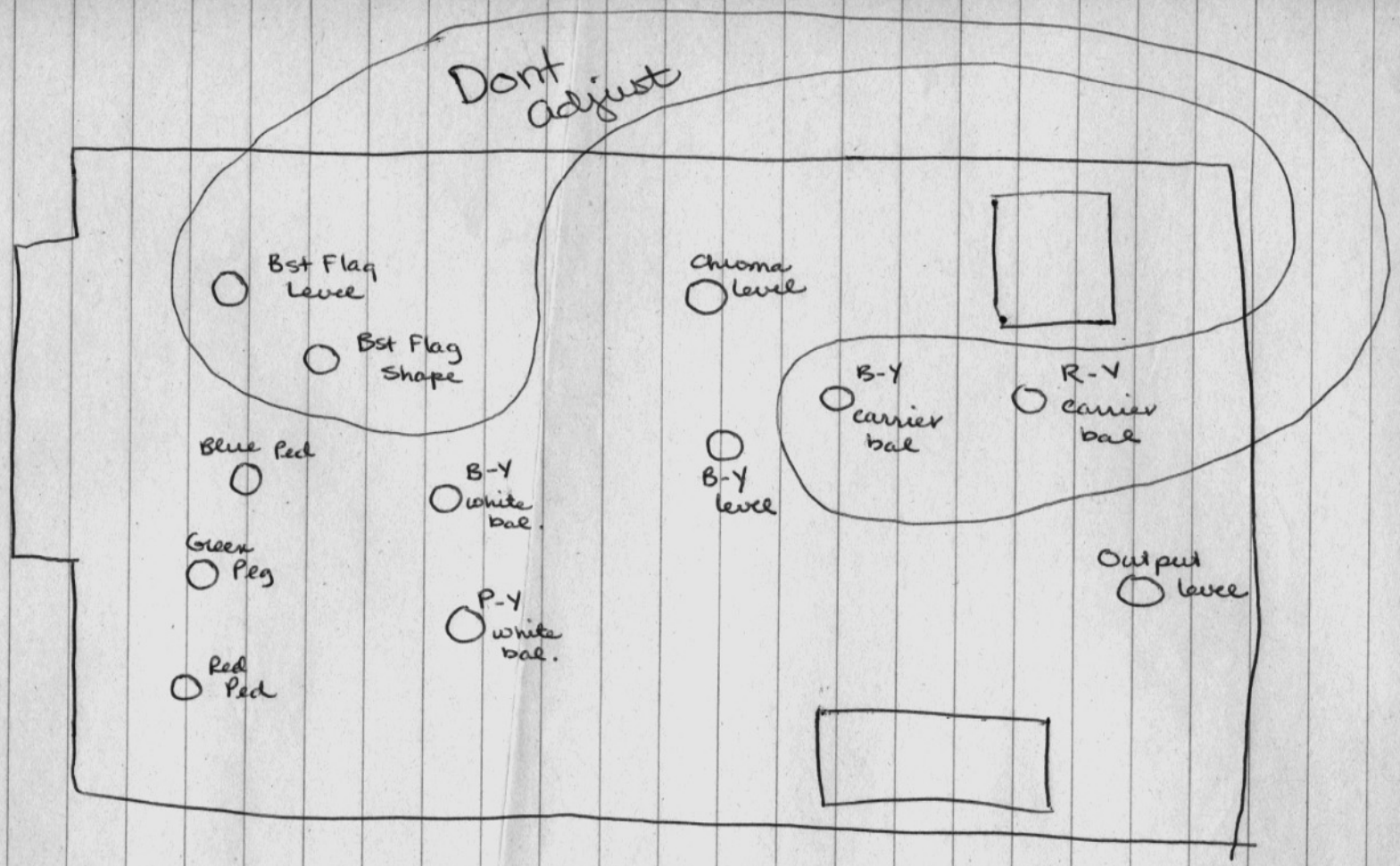
6-12. ENCODER

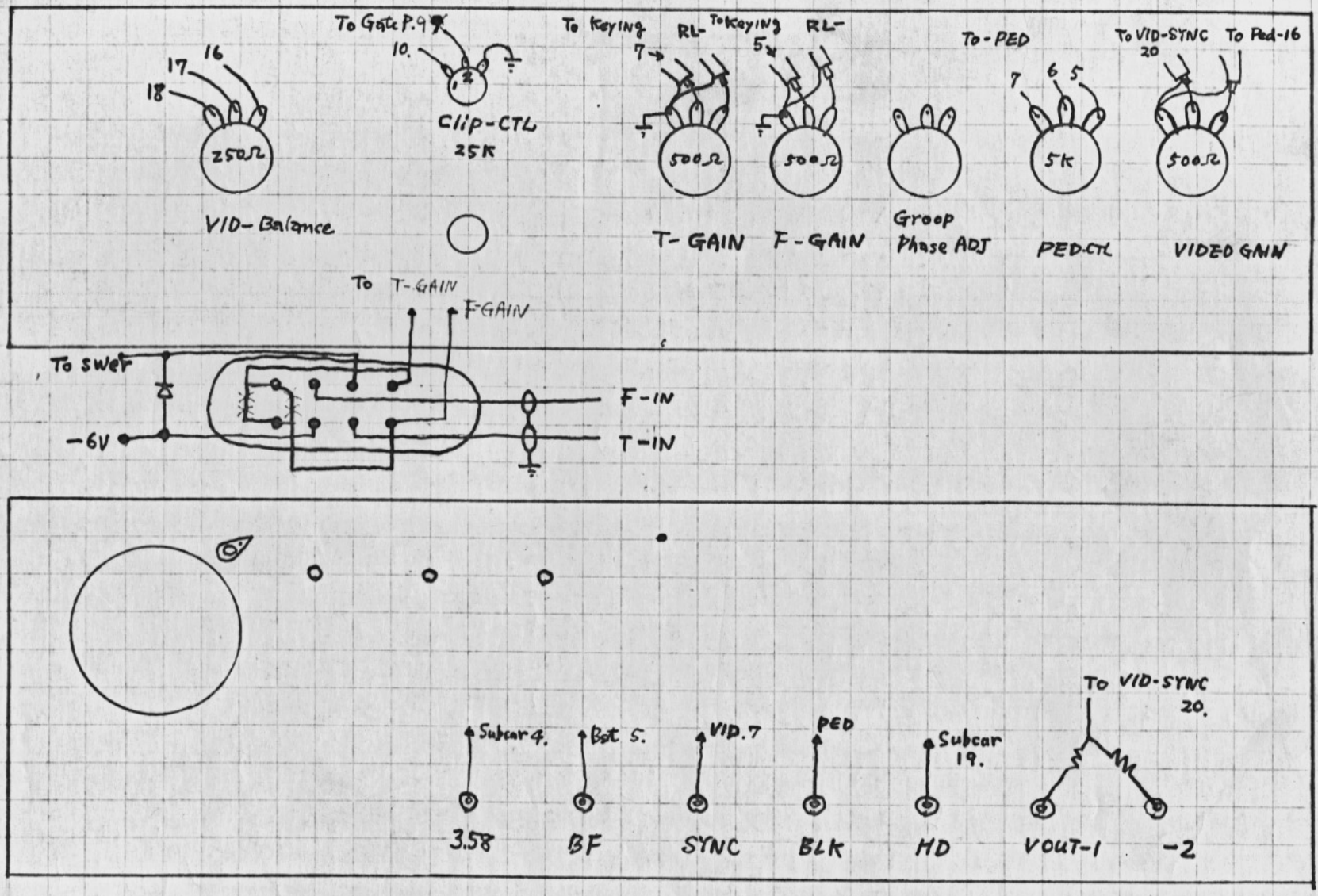


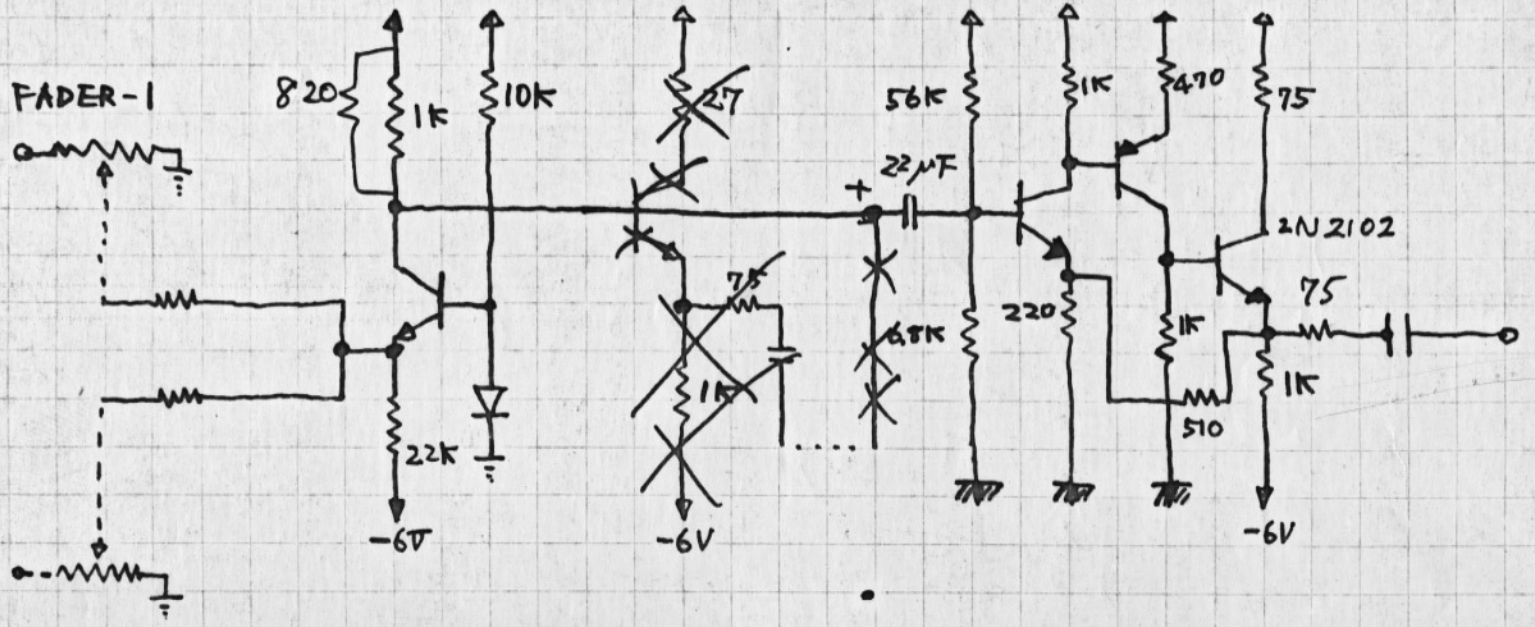
Notes:

1. Reference numbers are 7000 series.
2. All resistors in ohms, 5%, ¼W, unless otherwise noted.
3. All capacitors in μF unless otherwise noted.
4. The blue arrow indicates d c voltage.

top


Dont adjust
Bst Flag Level
Bst Flag Shape
Chroma level
B-Y carrier bal
R-Y carrier bal
Blue Ped
B-Y white bal.
B-Y level
Green Peg
P-Y white bal.
Output level
Red Ped


To Gate P.9
10.
Clip-CTL
25K
To Keying
7
RL-
To Keying
5
RL-
To-PED
To VID-SYNC
20
To Ped-16
18
17
16
250Ω
500Ω
500Ω
7
6
5
5K
500Ω
VID-Balance
T-GAIN
F-GAIN
Group
Phase ADJ
PED-CTL
VIDEO GNN
To T-GAIN
F-GAIN
To SWEP
-6V
F-IN
T-IN
Subcar 4.
Bat 5.
VID.7
PED
Subcar
19.
To VID-SYNC
20.
3.58
BF
SYNC
BLK
HD
VOUT-1
-2

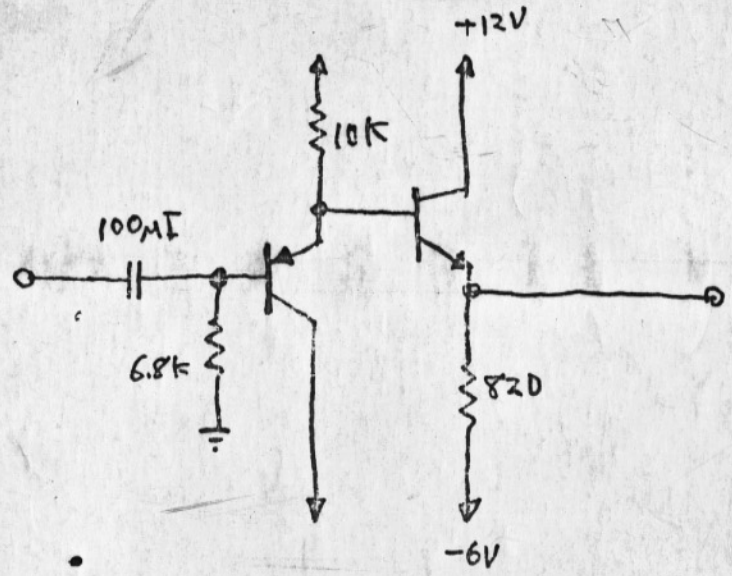


FADER -2

PNP ... 2N4402 or Equivalent

NPN ... 2N4400 " "

MIXER - FADER OUTPUT AMP



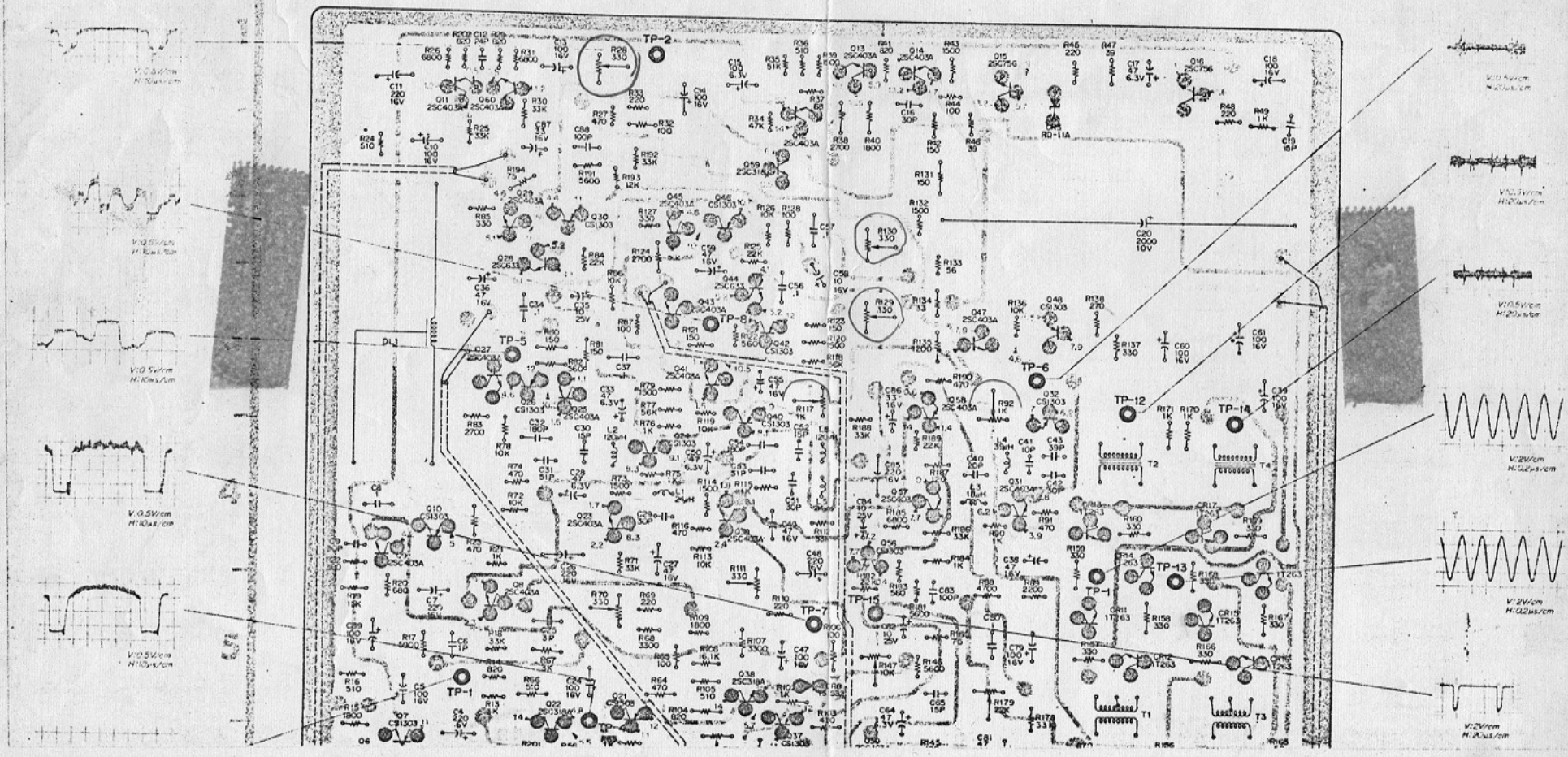
SAME AS Above

SAME AS Above

SWITCHER INPUT
EMITTER
Follower

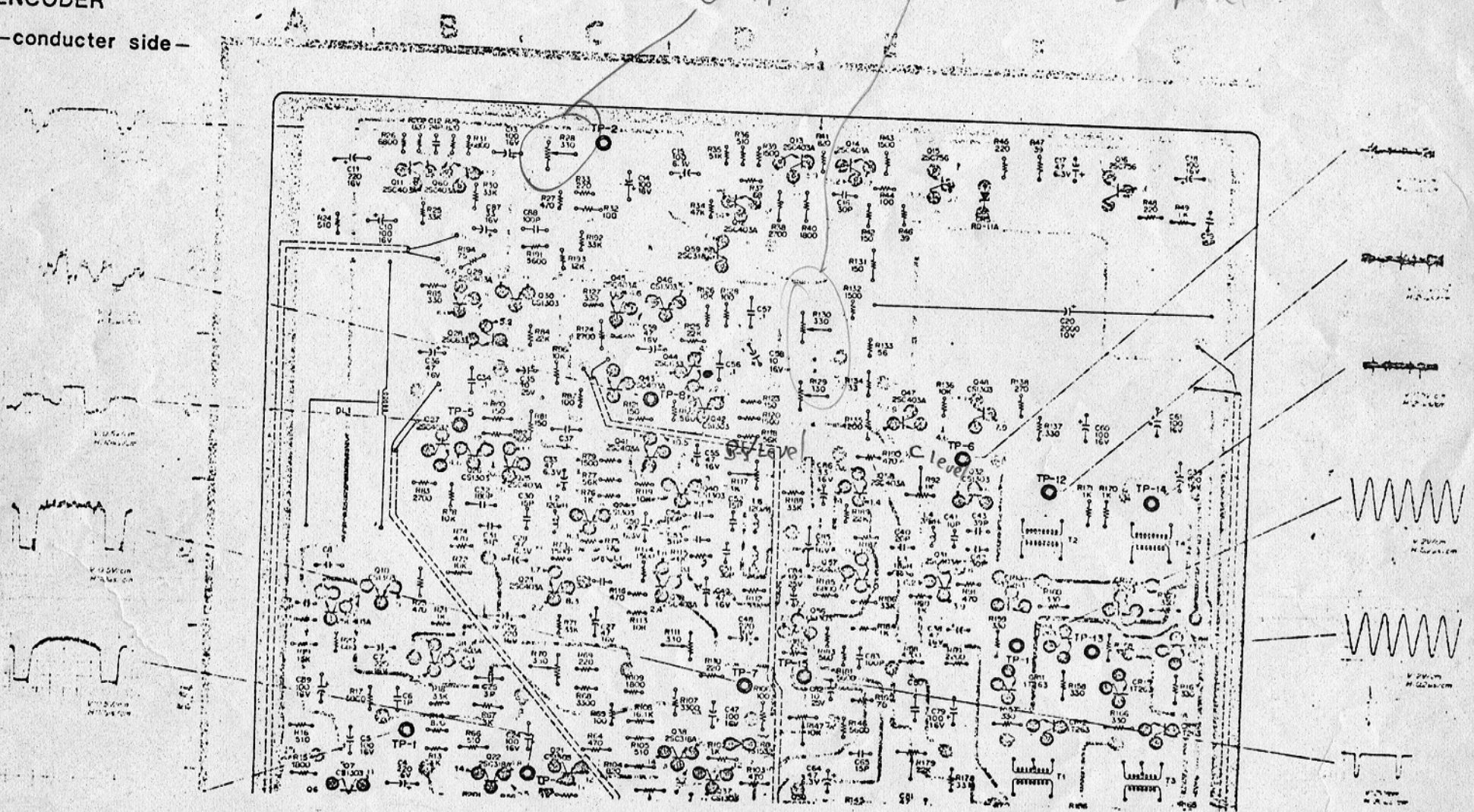
. ENCODER

—conducter side—

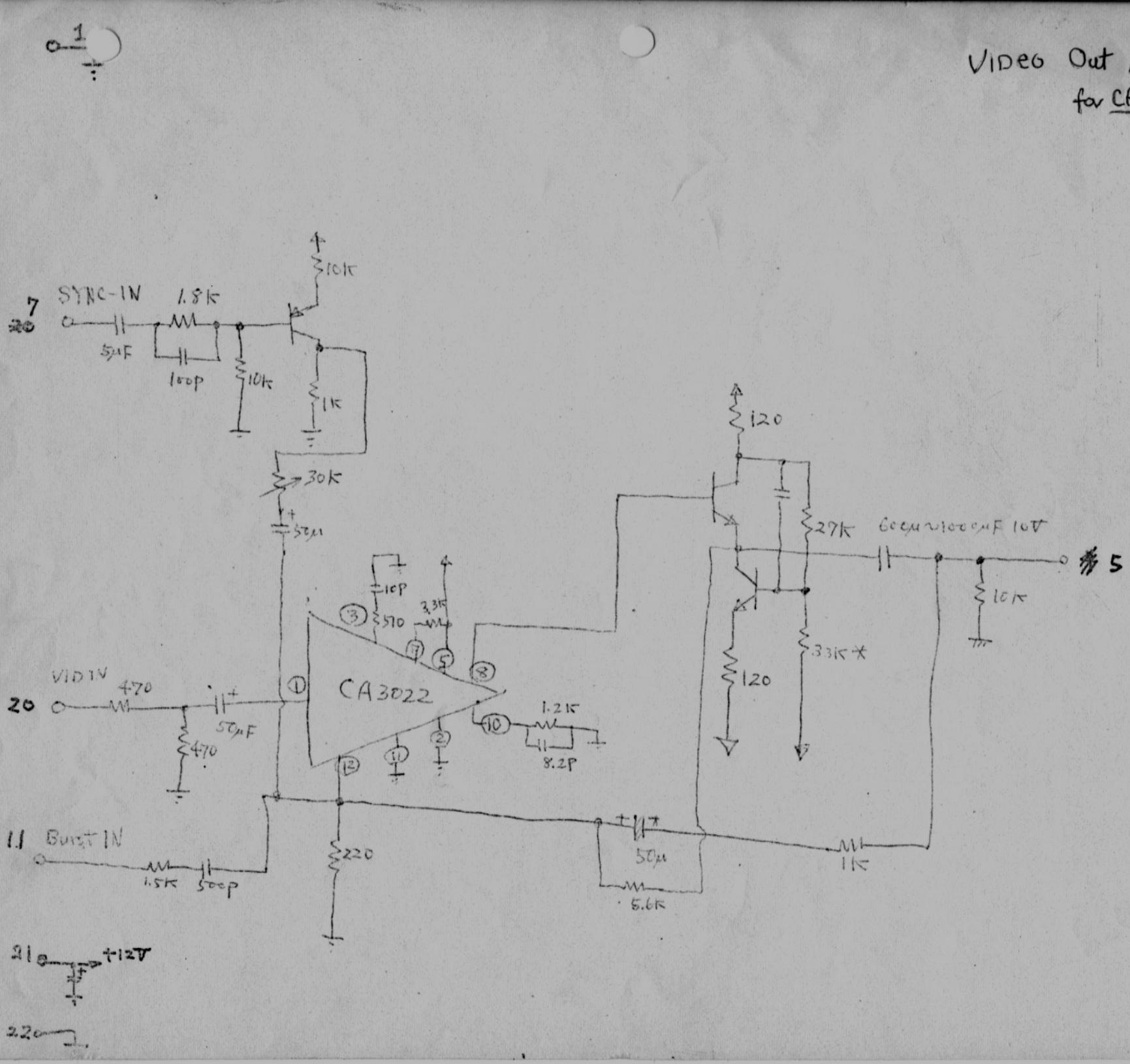


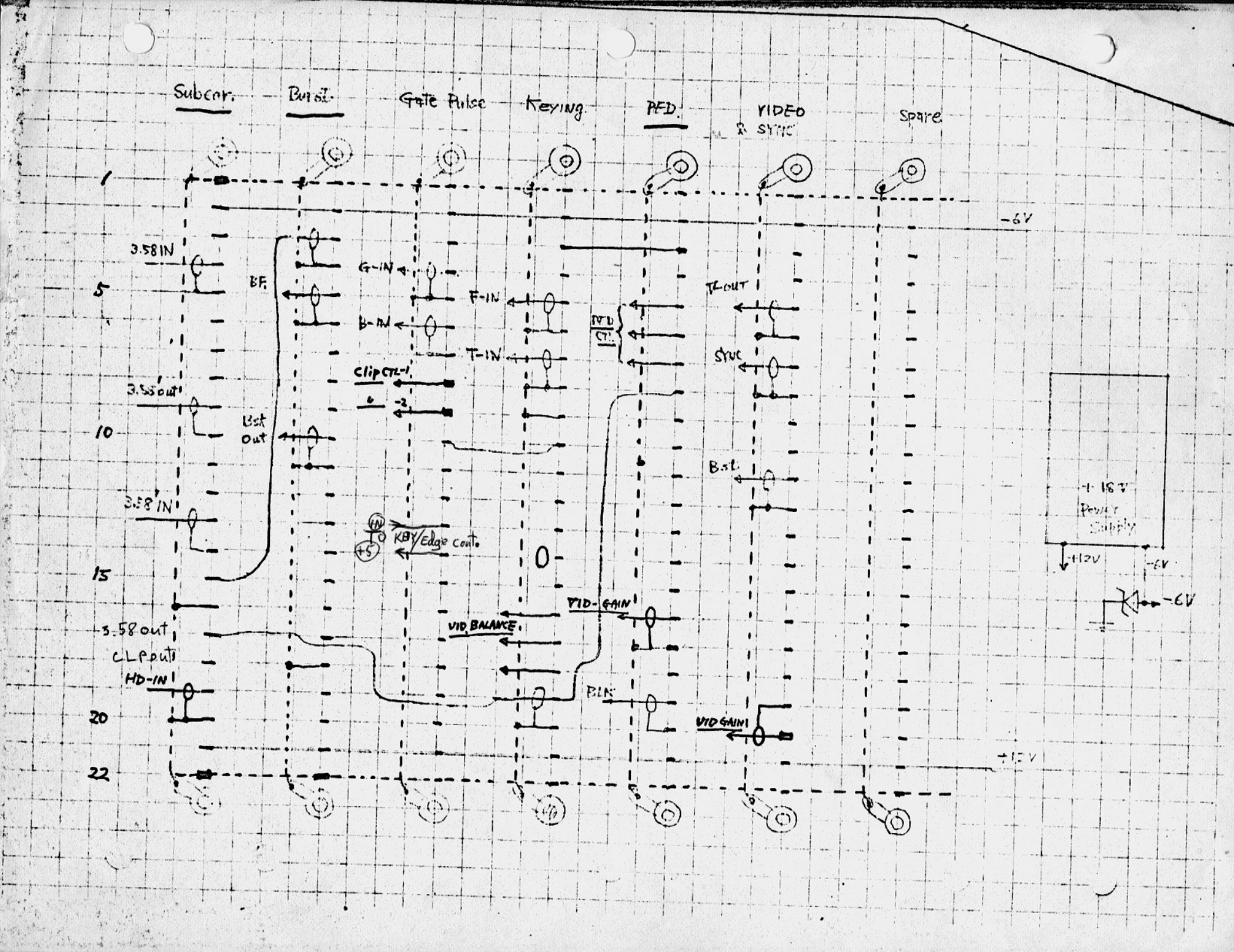
. ENCODER

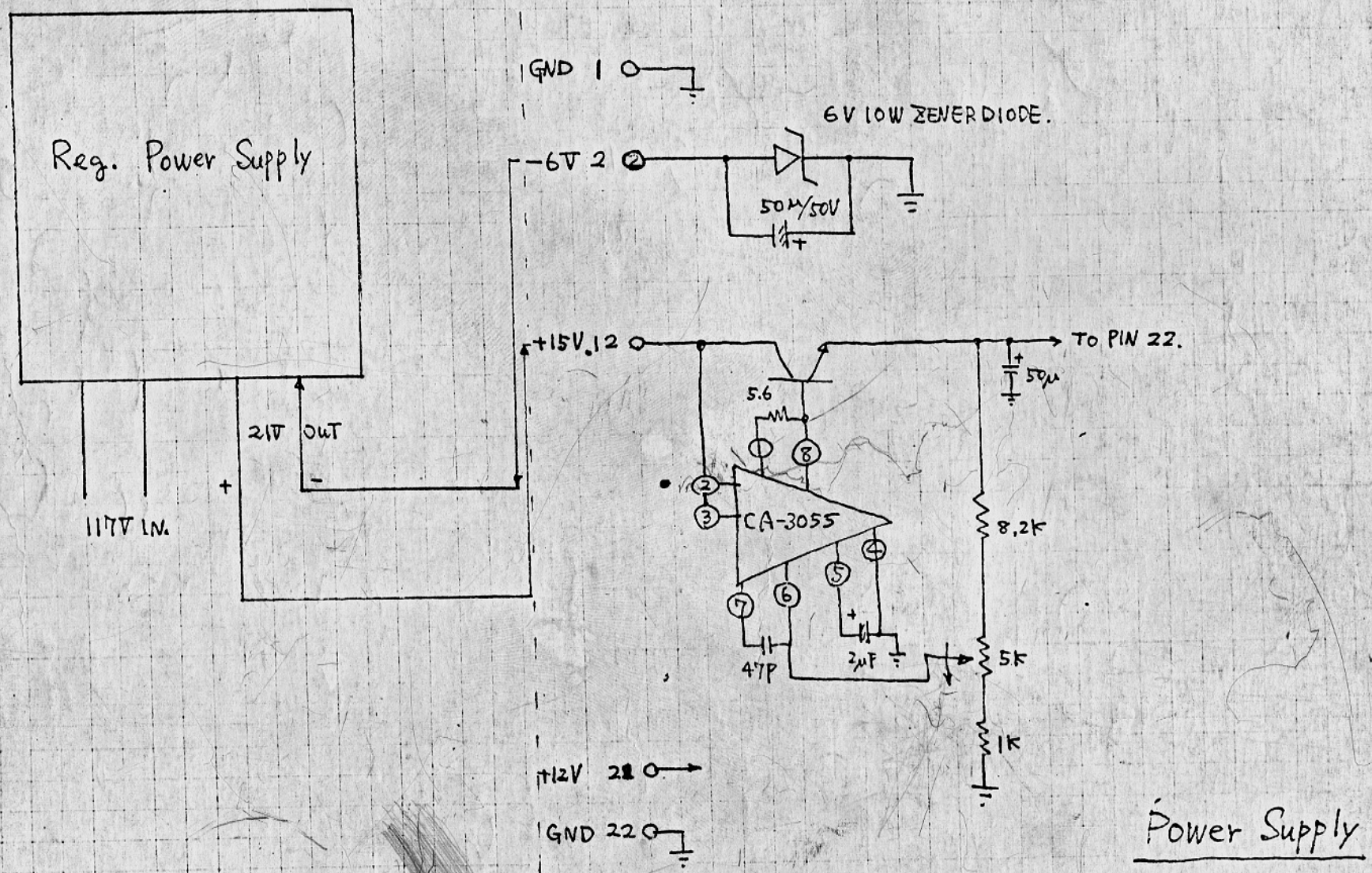
—conducter side—



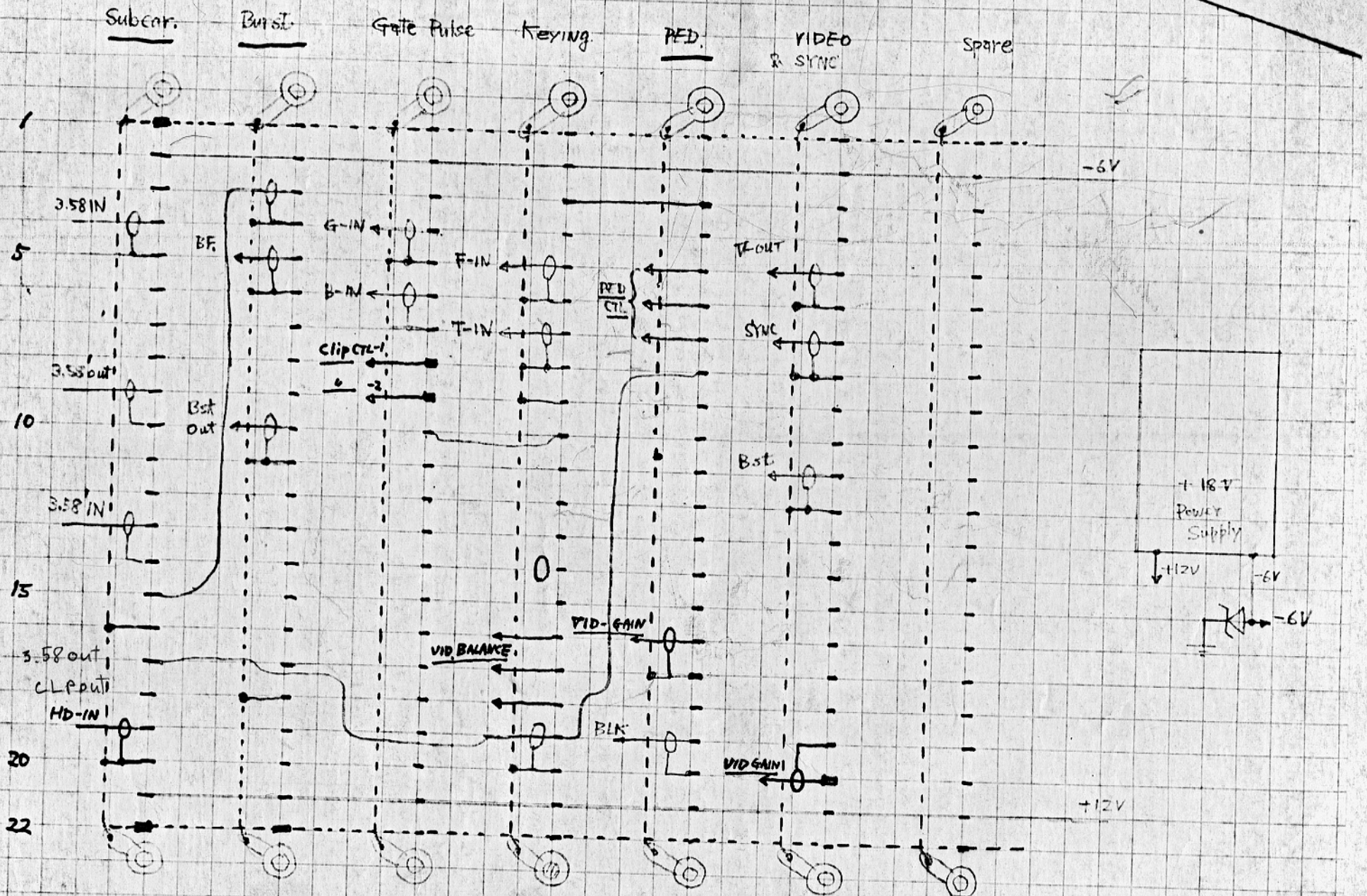
VIDEO Out AMP
for CROMA KEY

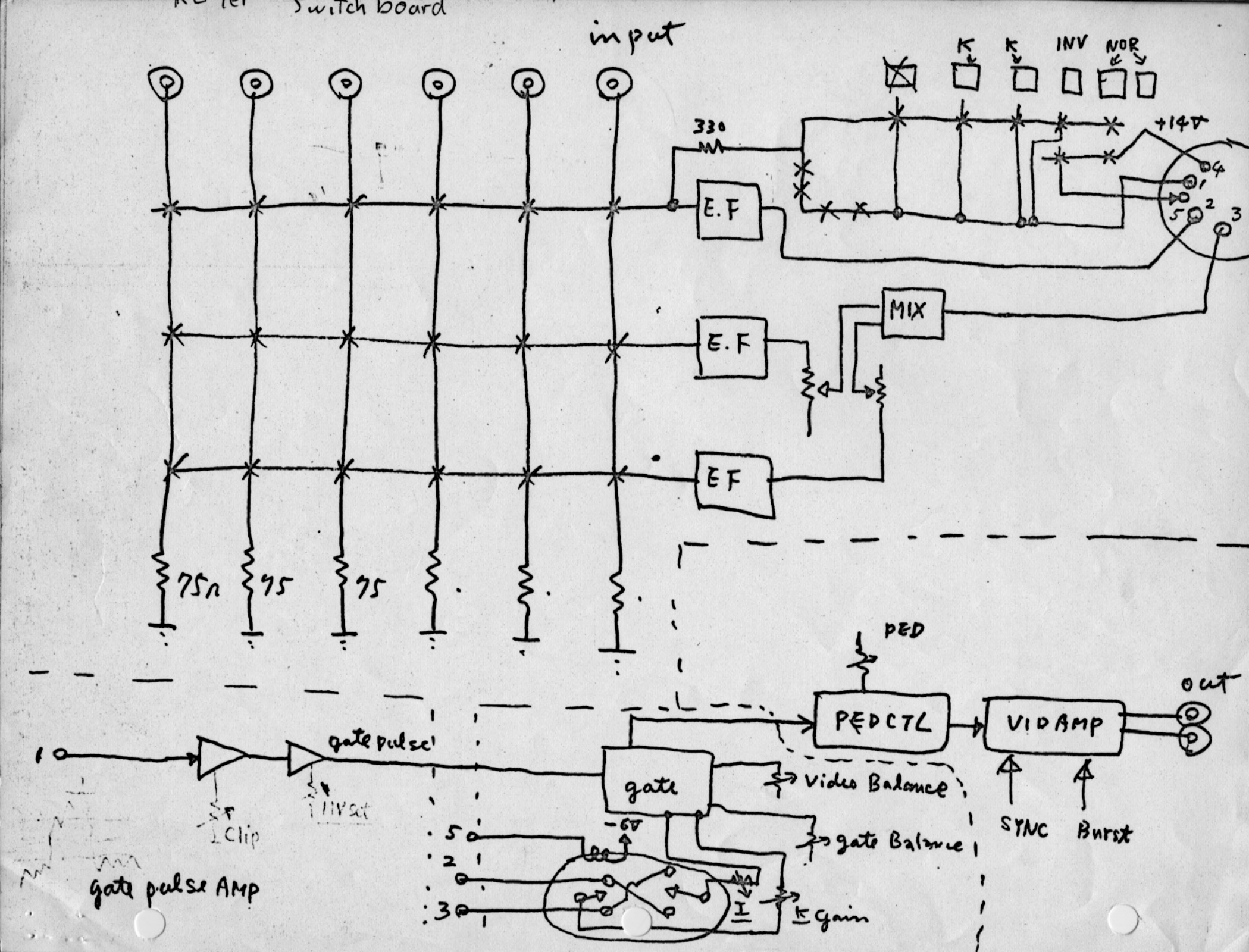


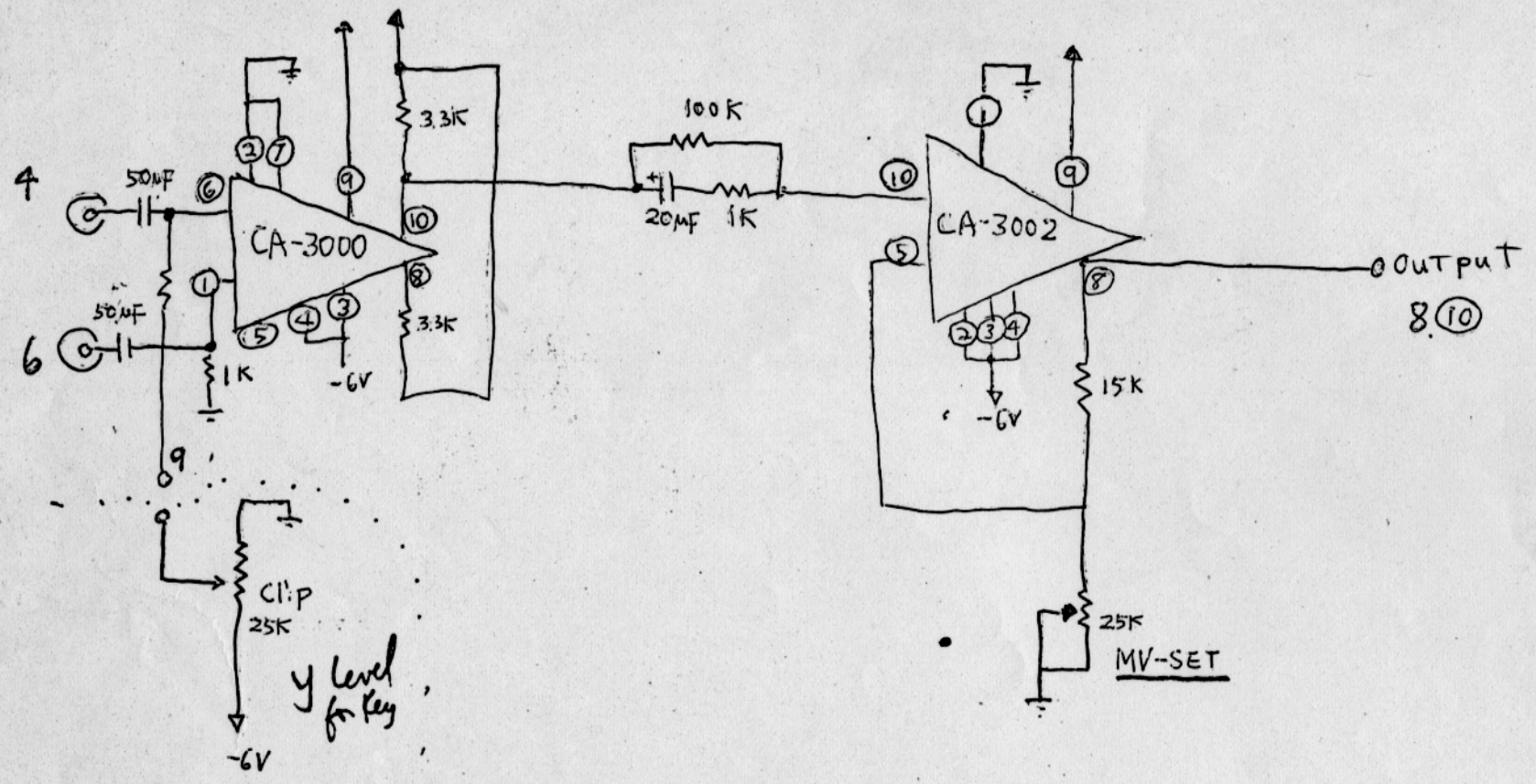

Subcar.
Burst
Gate Pulse
Keying
PED.
VIDEO & SYNC
Spare
1
5
10
15
20
22
-6V
+12V
3.58 IN
3.58 out
BF.
Burst out
3.58 IN
3.58 out
CL Pout
HD-IN
G-IN
B-IN
F-IN
T-IN
Clip CTL-1
-2
KEY/Edge cont.
+5
VID BALANCE
VID GAIN
PED CT.
TL-OUT
SYNC
Burst
VID GAIN
BLK
+18V Power Supply
+12V
-6V
-6V



Power Supply


Subcar.
Burst
Gate Pulse
Keying
RED.
VIDEO & SYNC
Spare
1
5
10
15
20
22
-6V
+12V
3.58 IN
3.58 out
BF.
Bst Out
3.58 IN
3.58 out
CL Pout
HD-IN
G-IN
B-IN
F-IN
T-IN
Clip CTL-1
-2
RED CTL
VID. BALANCE.
VID-GAIN
BLK
T-OUT
SYNC
Bst.
VID GAIN
+18V Power Supply
+12V
-6V
-6V

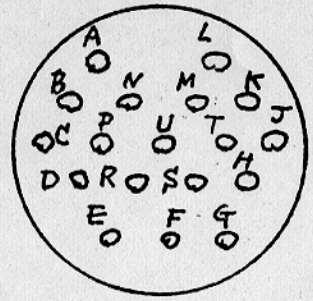

Switch board
input
K
K
INV
NOR
330
+14V
E.F
E.F
EF
MIX
75Ω
75
75
PED
out
PEDCTL
VIDAMP
gate pulse
gate
Video Balance
gate Balance
Clip
gate pulse AMP
-6V
SYNC
Burst
I
gain

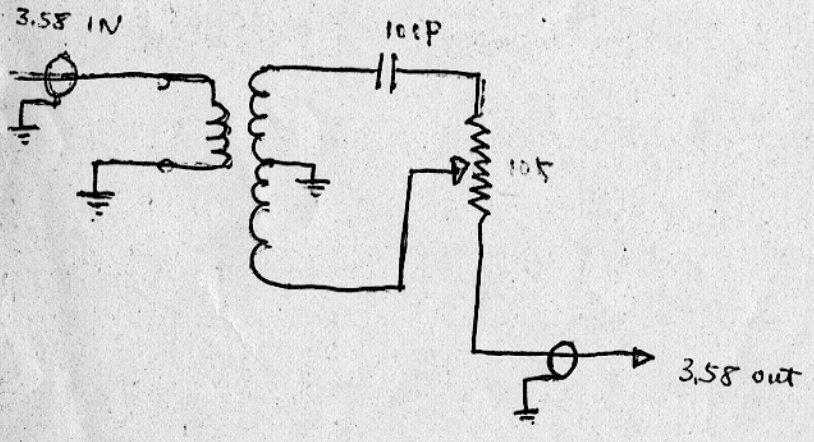


GATE Pulse Amp

CTC PANEL

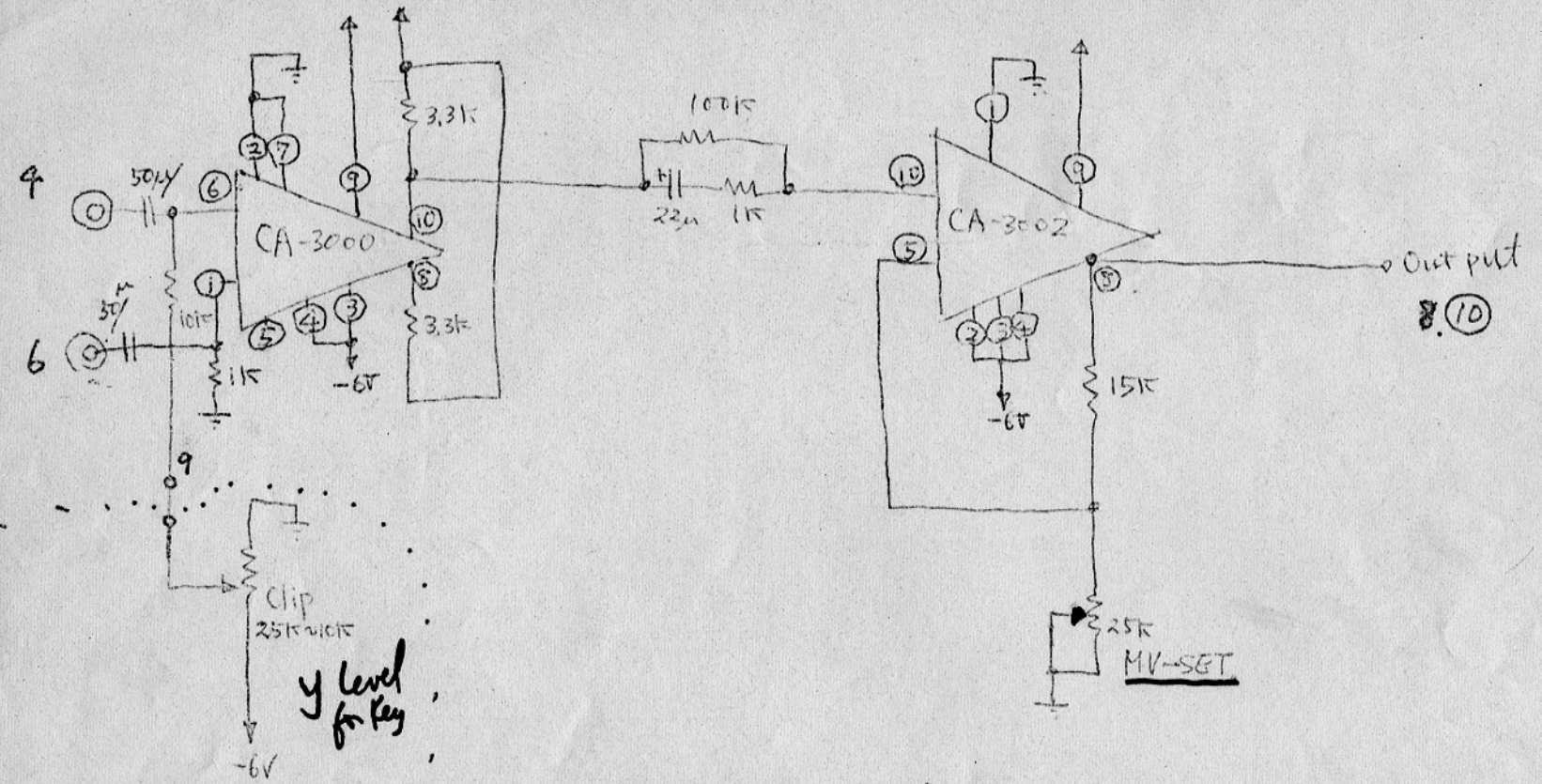
A	+ 12V
B	- 6V
P	GND
C	Rout
E	Coax SEL VID
H	F out
L	T out
F	GND
K	"





Phase CTL.

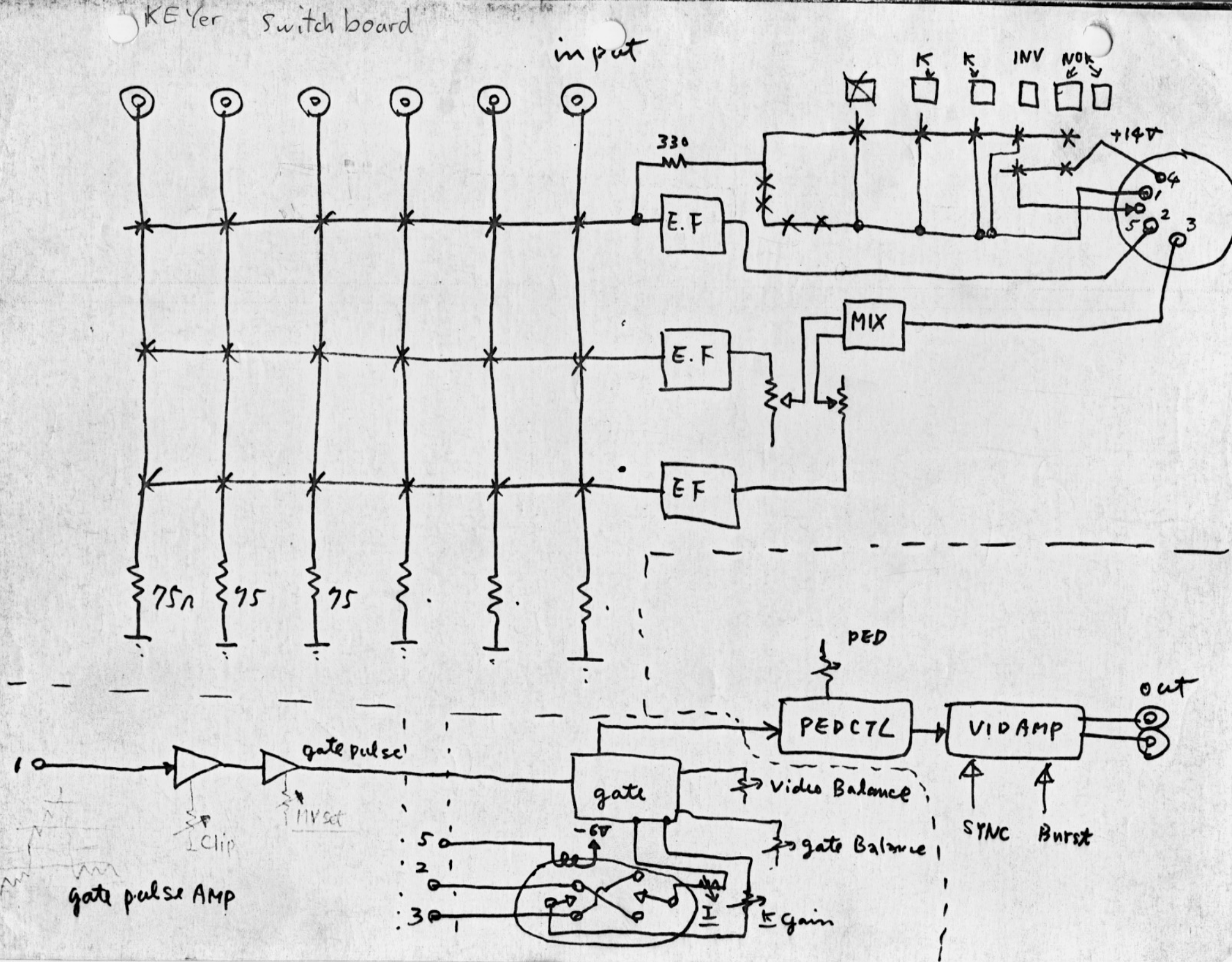
CONNECTOR Diagram

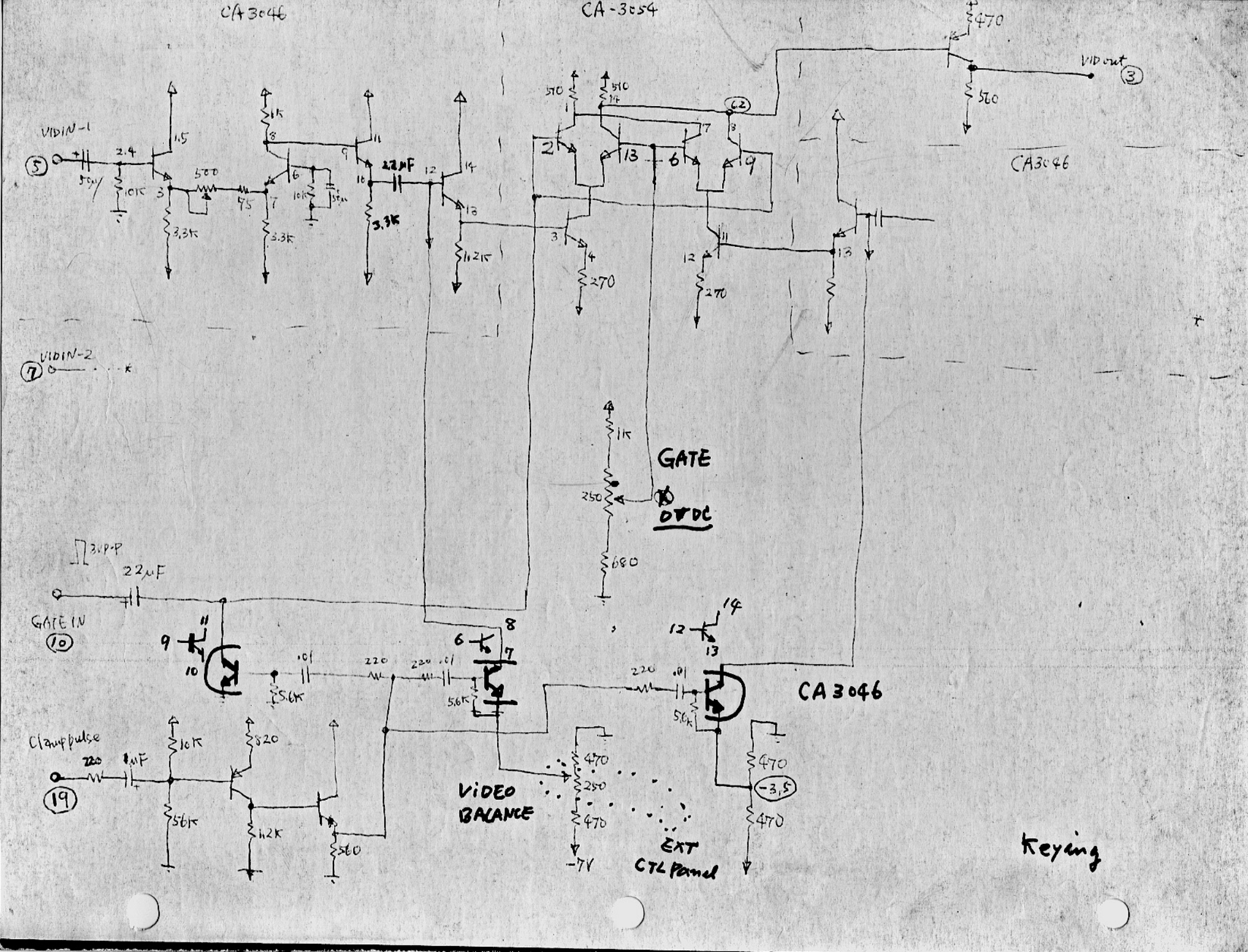


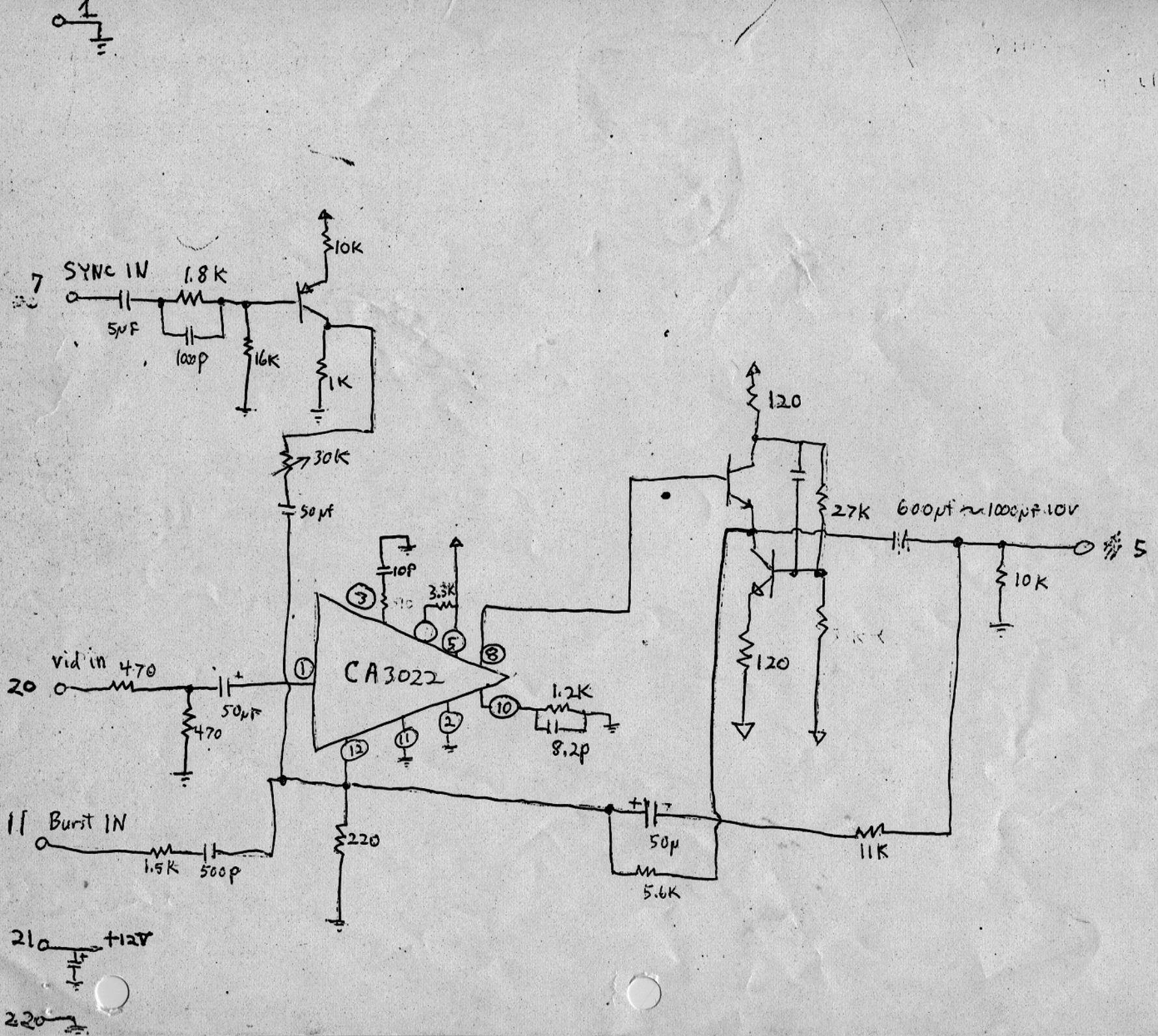
GATE P. AMP

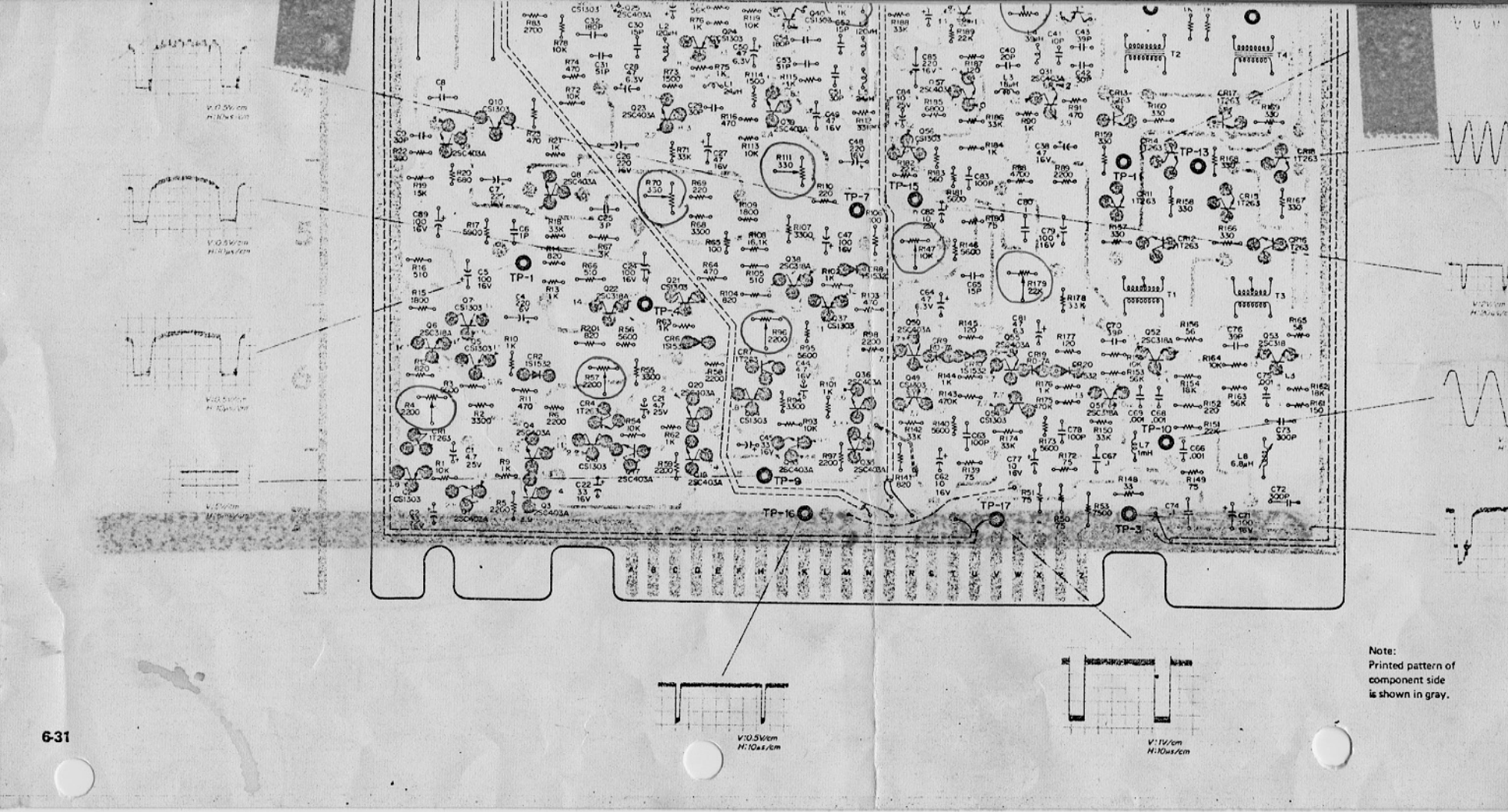
CTC PANEL

KEYer Switch board

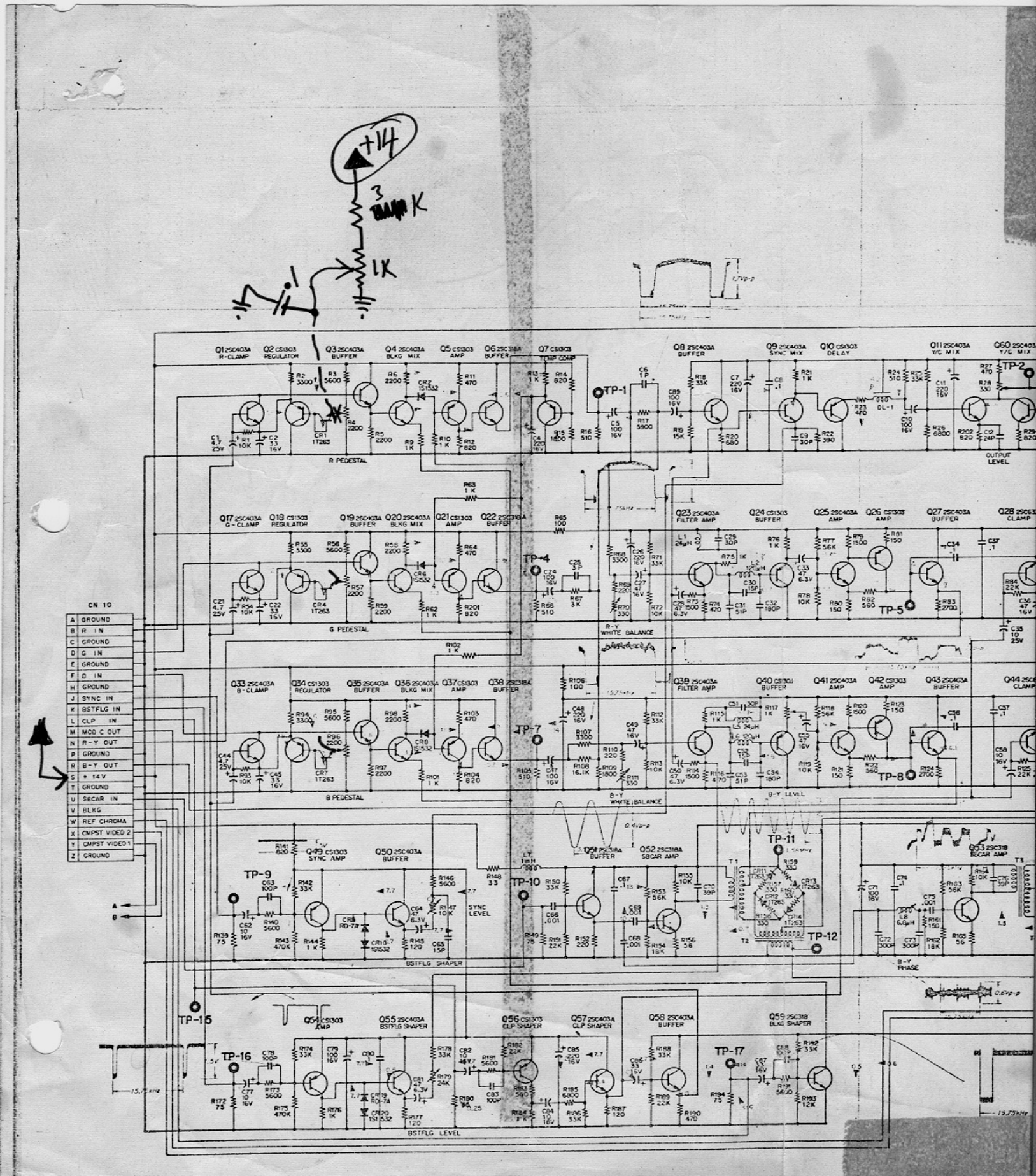



CA3046
CA-3054
VIDIN-1
VIDIN-2
VID out
CA3046
GATE
DC
GATE IN
Clamp pulse
VIDEO BALANCE
EXT CTL Panel
CA3046
Keying

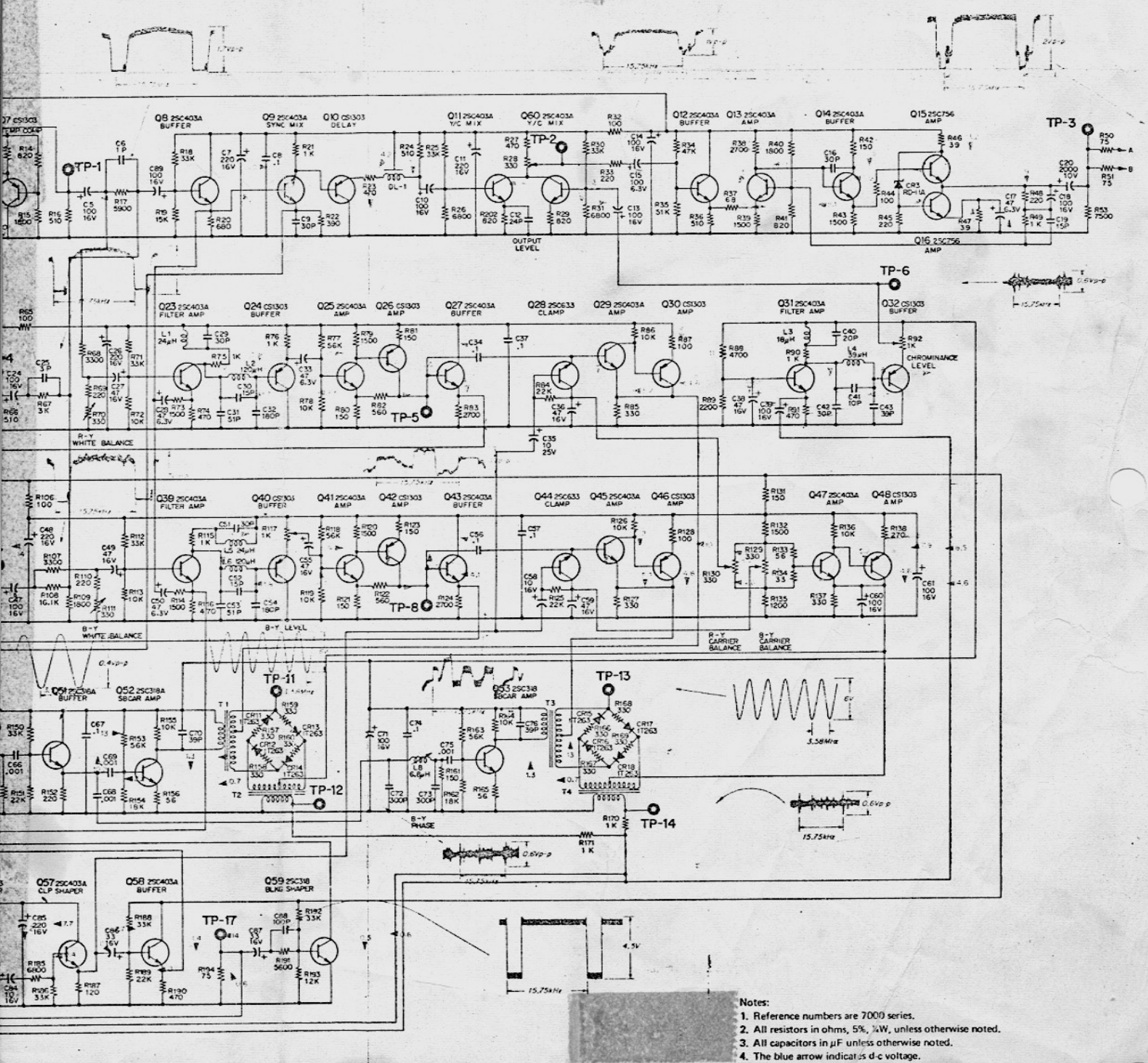

1
7
SYNC IN
1.8K
10K
5μF
1000p
16K
1K
30K
50μf
120
27K
600μf ≈ 1000μf 10V
5
10K
10p
3.3K
3
5
8
vid in
20
470
1
CA3022
50μF
470
10
1.2K
8.2p
2
11
12
11
Burst IN
1.5K
500p
220
50μ
11K
5.6K
21
+12V
220



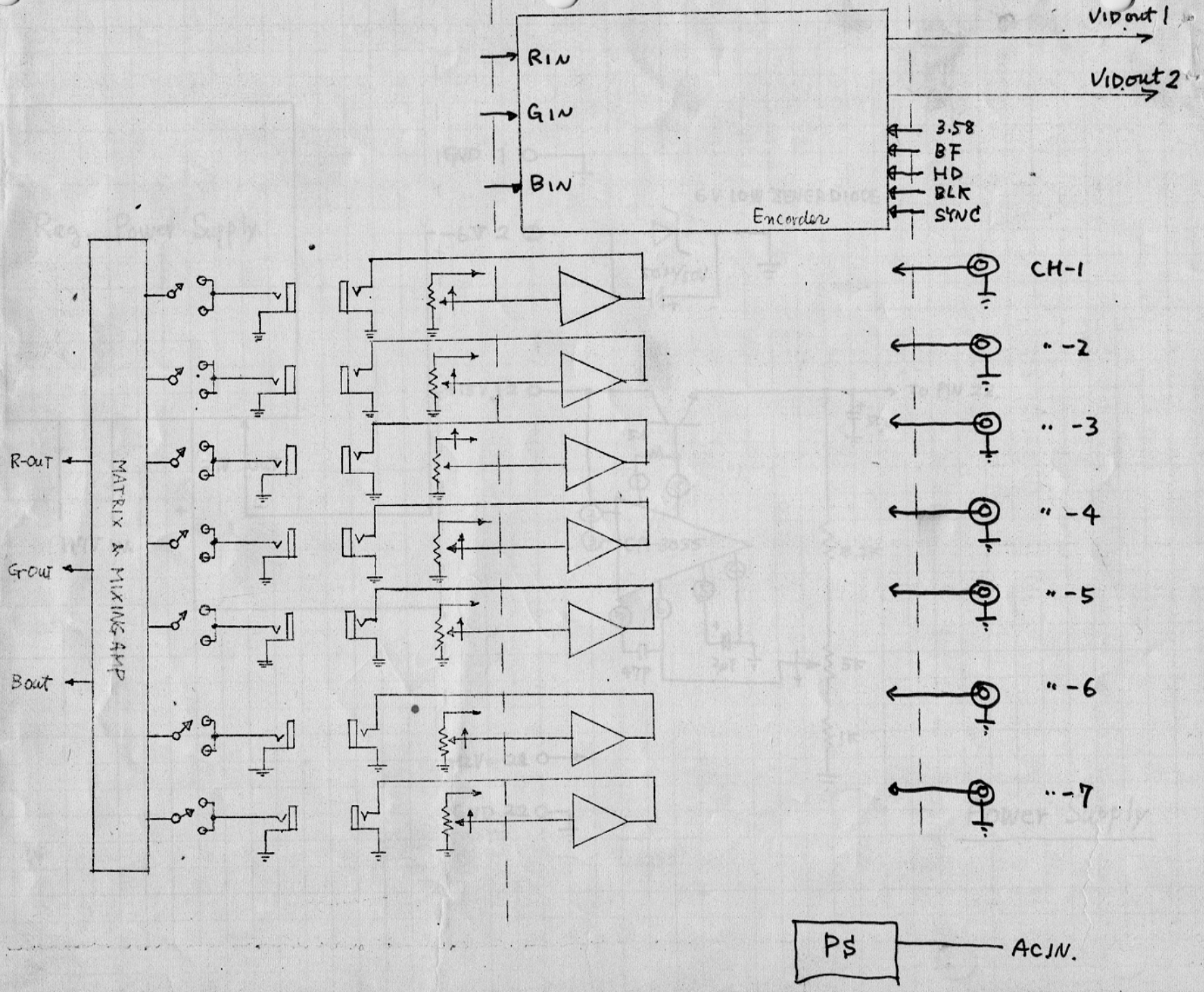
Note:
Printed pattern of component side is shown in gray.

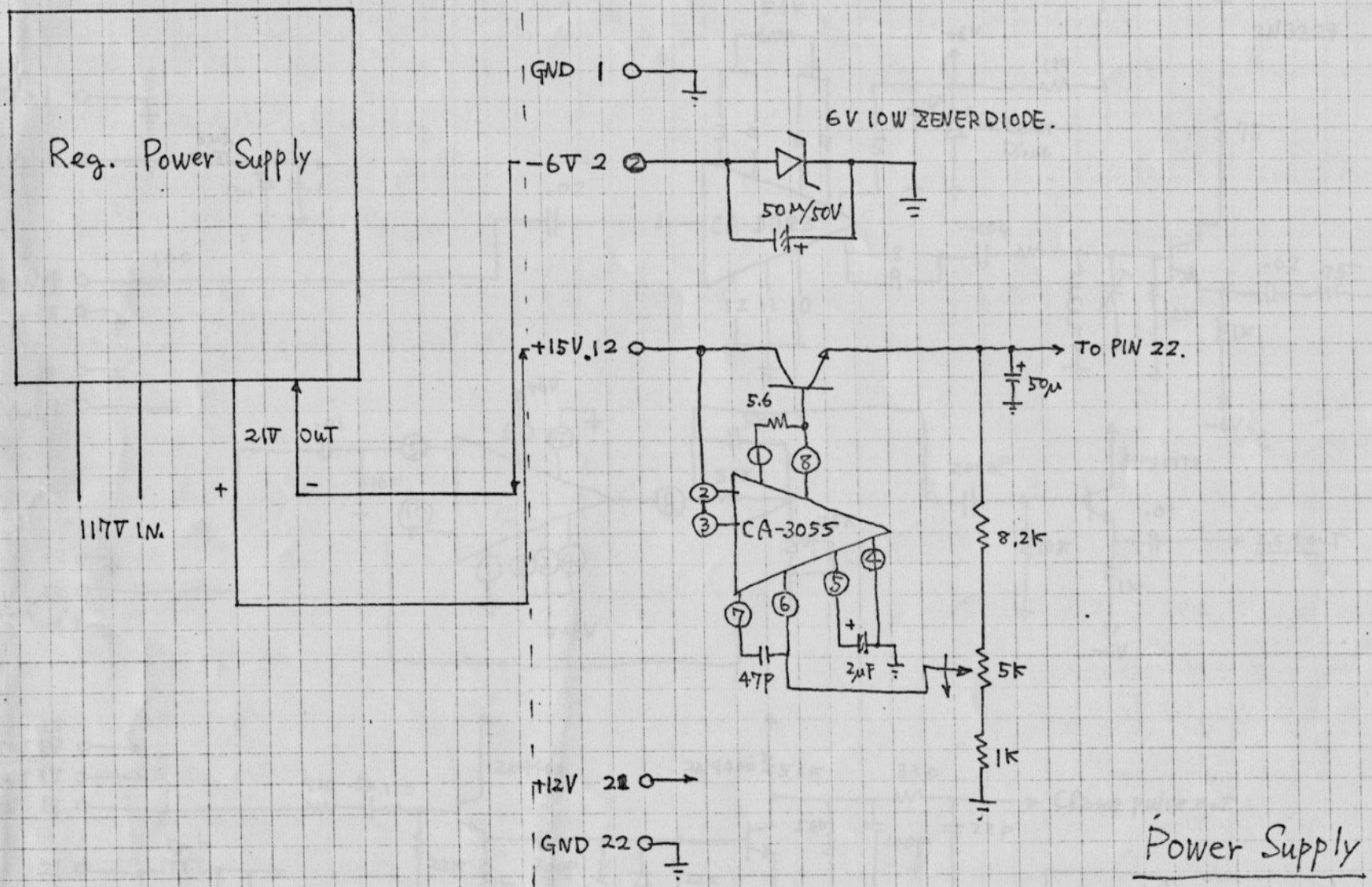

+14
3
K
1K
CN 10
A GROUND
B R IN
C GROUND
D G IN
E GROUND
F B IN
H GROUND
J SYNC IN
K BSTFLG IN
L CLP IN
M MOD C OUT
N R-Y OUT
P GROUND
R B-Y OUT
S + 14V
T GROUND
U SBCAR IN
V BLKG
W REF CHROMA
X CMPST VIDEO 2
Y CMPST VIDEO 1
Z GROUND
TP-1
TP-2
TP-4
TP-5
TP-7
TP-8
TP-9
TP-10
TP-11
TP-12
TP-15
TP-16
TP-17
R PEDESTAL
G PEDESTAL
B PEDESTAL
R-Y WHITE BALANCE
B-Y WHITE BALANCE
B-Y LEVEL
B-Y PHASE
OUTPUT LEVEL
SYNC LEVEL
BSTFLG SHAPER
BSTFLG LEVEL
DL-1
Q1 2SC403A R-CLAMP
Q2 CS1303 REGULATOR
Q3 2SC403A BUFFER
Q4 2SC403A BLKG MIX
Q5 CS1303 AMP
Q6 2SC388A BUFFER
Q7 CS1303 COMP COMP
Q8 2SC403A BUFFER
Q9 2SC403A SYNC MIX
Q10 CS1303 DELAY
Q11 2SC403A Y/C MIX
Q60 2SC403
Q17 2SC403A G-CLAMP
Q18 CS1303 REGULATOR
Q19 2SC403A BUFFER
Q20 2SC403A BLKG MIX
Q21 CS1303 AMP
Q22 2SC388A BUFFER
Q23 2SC403A FILTER AMP
Q24 CS1303 BUFFER
Q25 2SC403A AMP
Q26 CS1303 AMP
Q27 2SC403A BUFFER
Q28 2SC63 CLAMP
Q33 2SC403A B-CLAMP
Q34 CS1303 REGULATOR
Q35 2SC403A BUFFER
Q36 2SC403A BLKG MIX
Q37 CS1303 AMP
Q38 2SC388A BUFFER
Q39 2SC403A FILTER AMP
Q40 CS1303 BUFFER
Q41 2SC403A AMP
Q42 2SC403A AMP
Q43 2SC403A BUFFER
Q44 2SC
Q49 CS1303 SYNC AMP
Q50 2SC403A BUFFER
Q51 2SC388A BUFFER
Q52 2SC388A SBCAR AMP
Q53 2SC38 SBCAR AMP
Q54 CS1303 AMP
Q55 2SC403A BSTFLG SHAPER
Q56 CS1303 CLP SHAPER
Q57 2SC403A CLP SHAPER
Q58 2SC403A BUFFER
Q59 2SC38 BLKG SHAPER
0.4Vp-p
0.6Vp-p
1.7Vp-p
15.75kHz

6-12. ENCODER

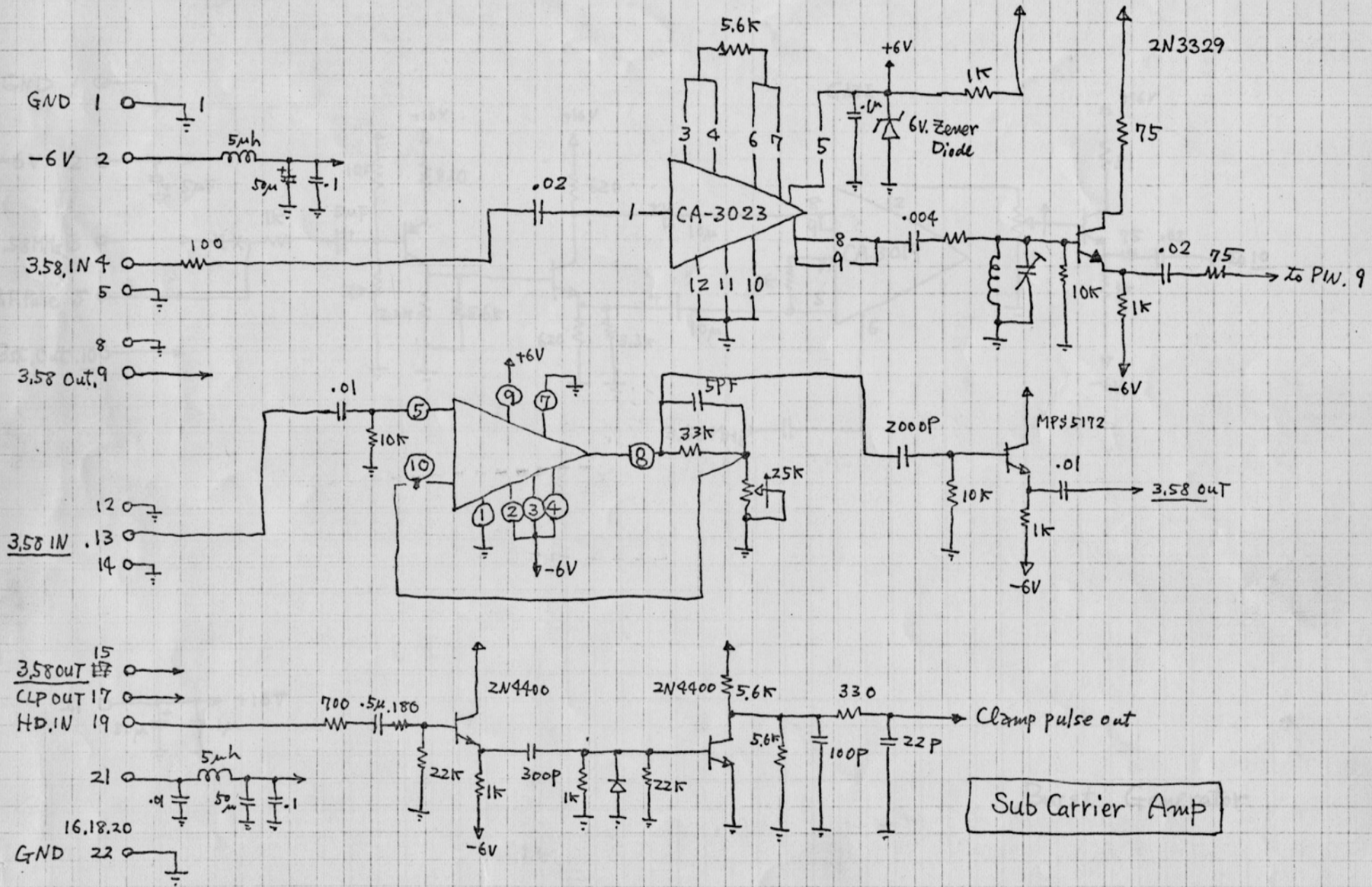


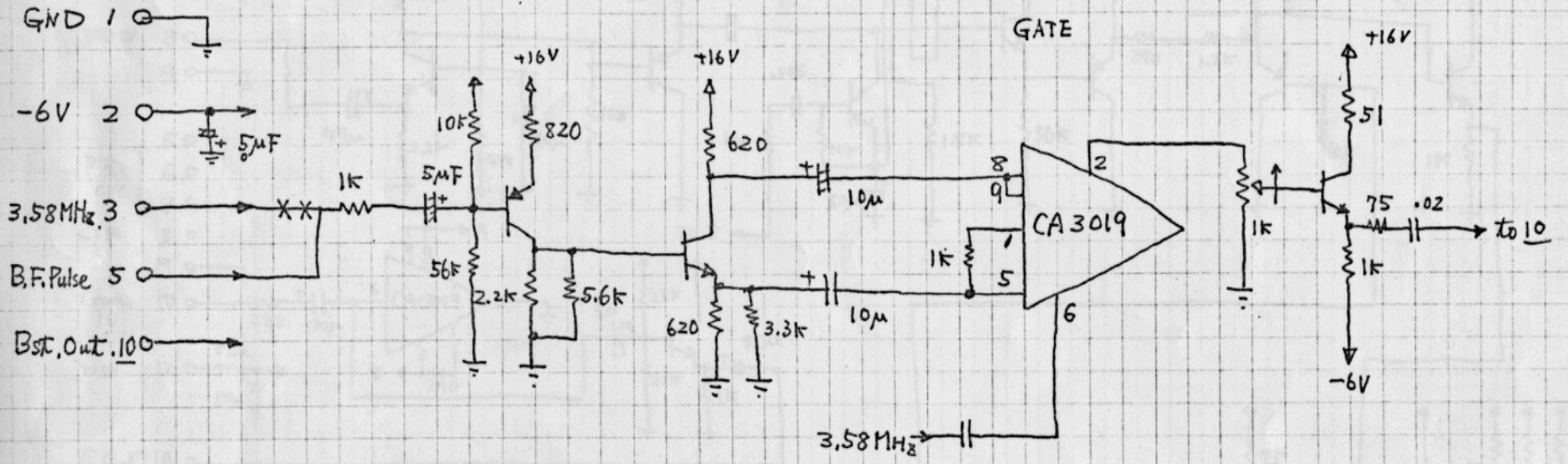
Notes:
1. Reference numbers are 7000 series.
2. All resistors in ohms, 5%, ¼W, unless otherwise noted.
3. All capacitors in μF unless otherwise noted.
4. The blue arrow indicates d-c voltage.

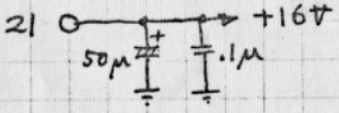

RIN
GIN
BIN
Encorder
VIDout 1
VIDout 2
3.58
BF
HD
BLK
SYNC
CH-1
" -2
" -3
" -4
" -5
" -6
" -7
R-OUT
GOUT
BOUT
MATRIX & MIXING AMP
PS
ACIN.



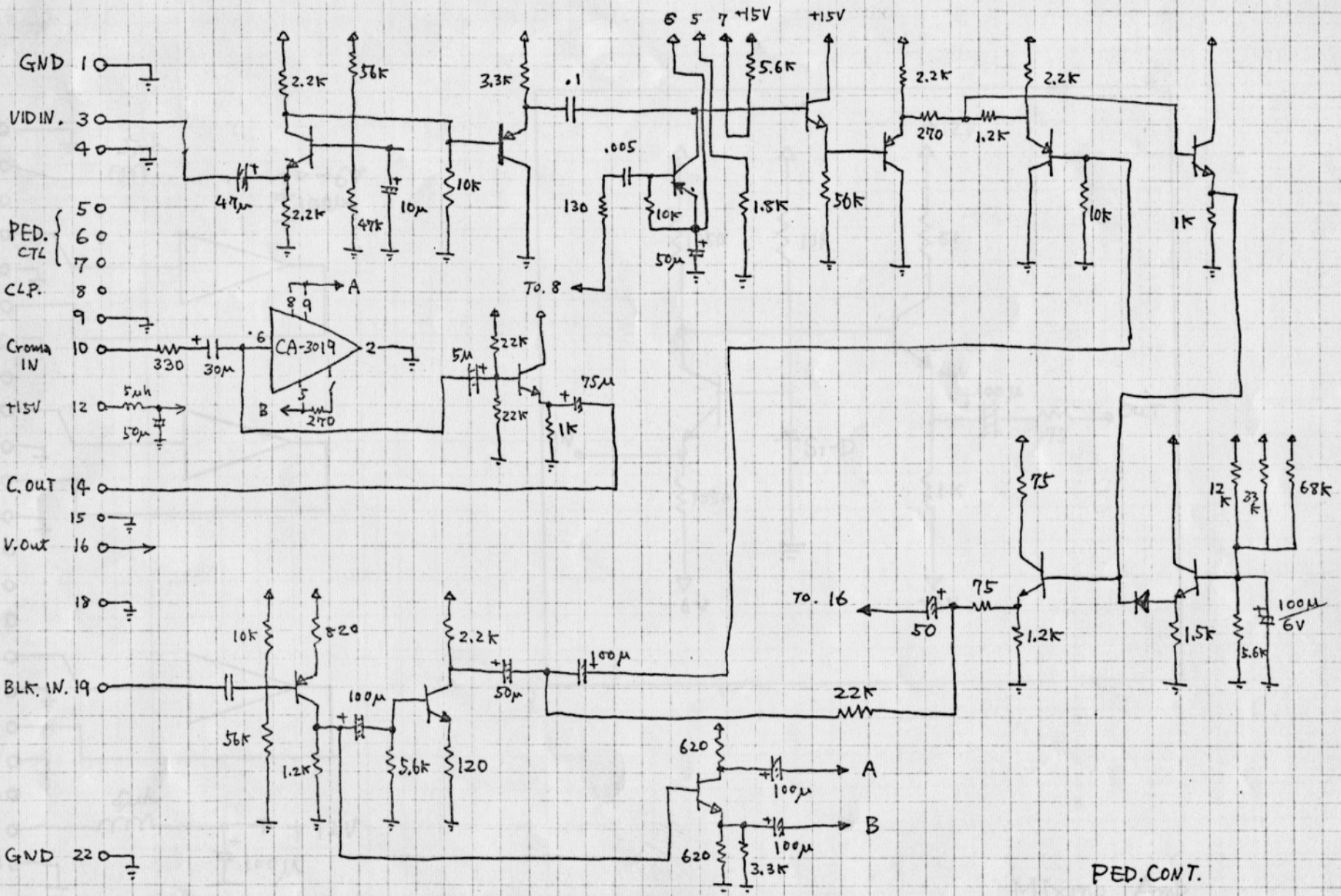
Power Supply

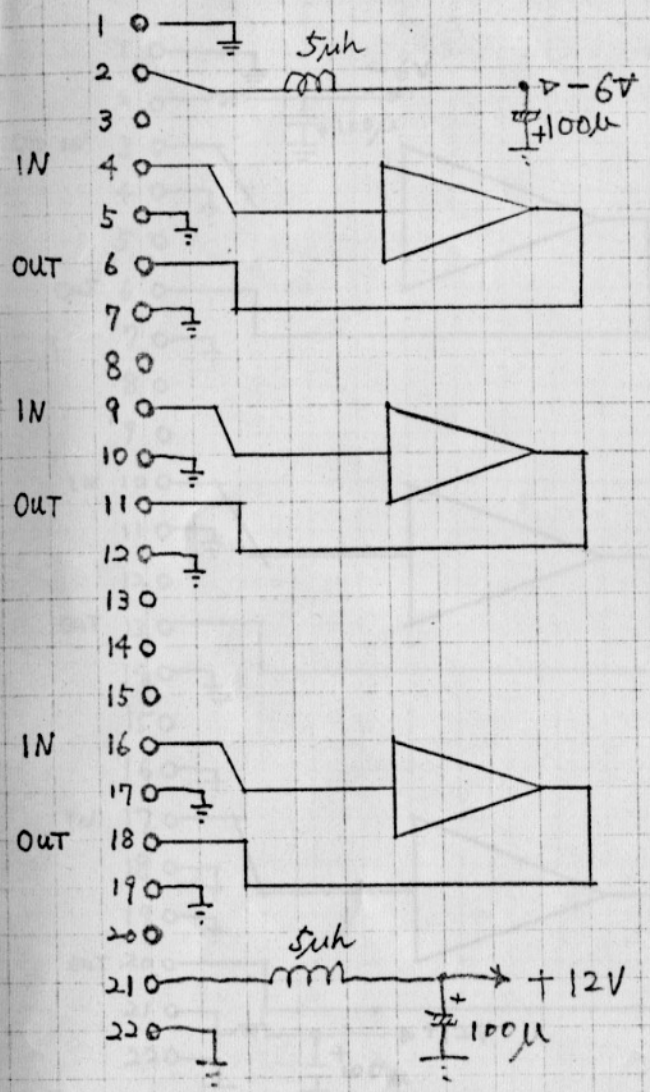

GND 1
-6V 2
3.58 IN 4
5
8
3.58 out 9
12
3.58 IN 13
14
3.58 OUT 15
CLP OUT 17
HD. IN 19
16.18.20
GND 22
5µh
50µ
.1
.02
100
CA-3023
5.6K
+6V
1K
6V. Zener Diode
2N3329
75
.004
10K
.02
to PIN. 9
-6V
.01
5PF
33K
25K
2000P
MPS5172
700
.5µ
.180
2N4400
22K
300P
330
100P
22P
Clamp pulse out
Sub Carrier Amp

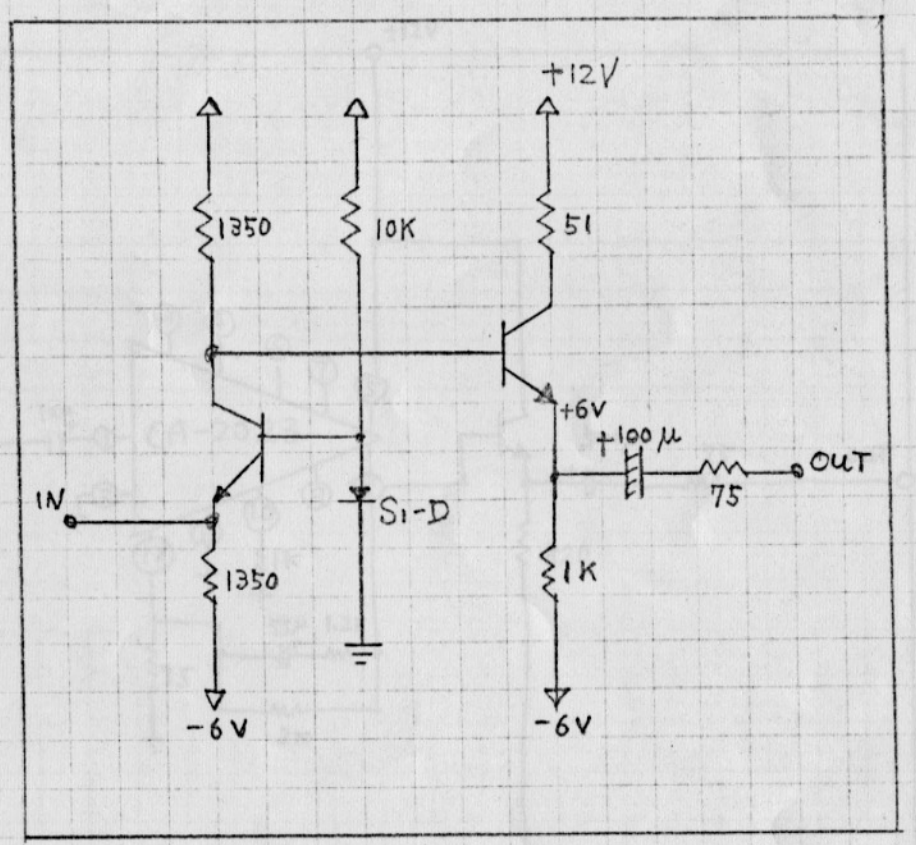




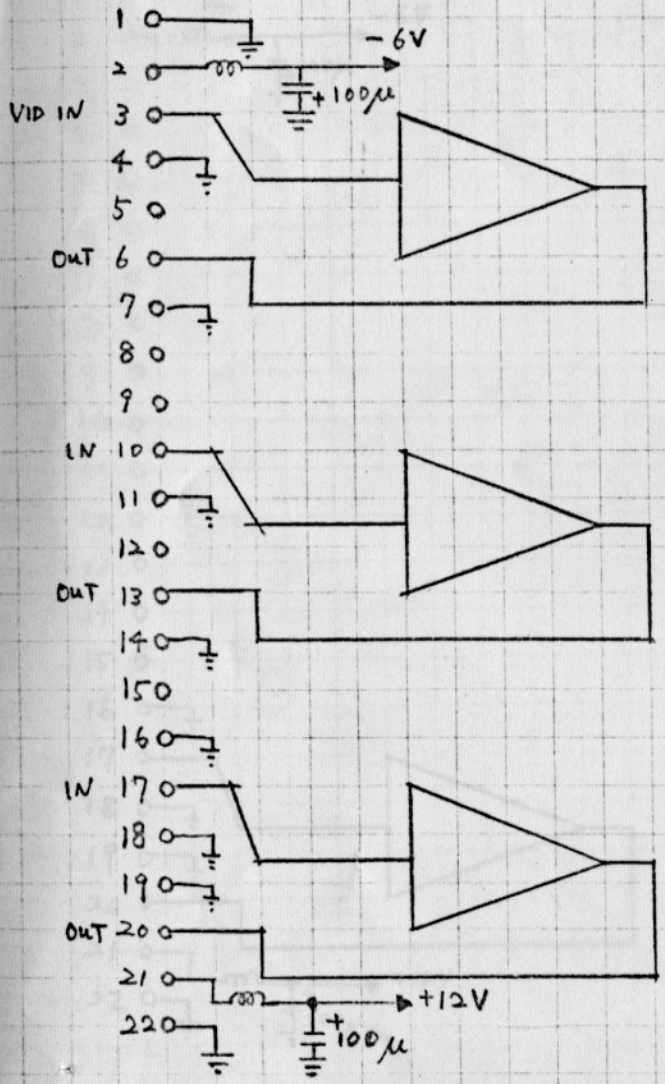
Burst Generator

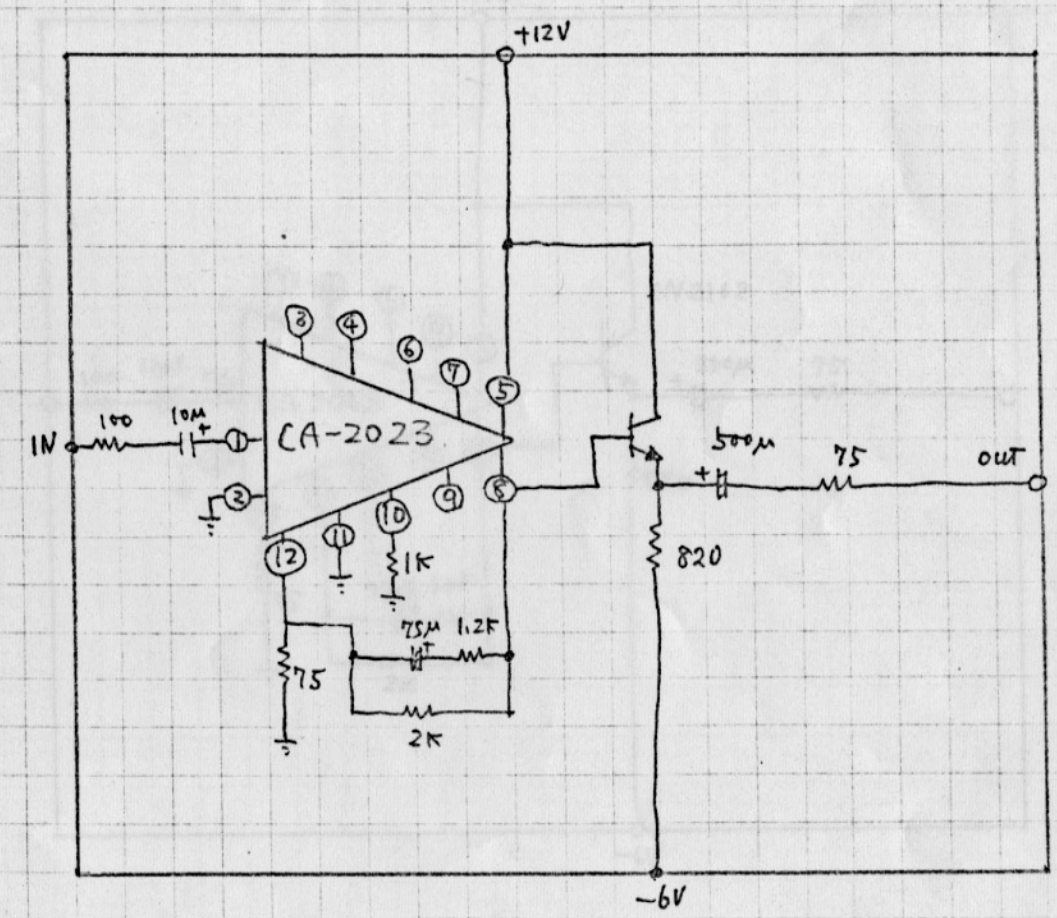

GND 1
VID IN. 3
4
PED. CTL 5 6 7
CLP. 8
9
Croma IN 10
+15V 12
C. OUT 14
15
V. Out 16
18
BLK. IN. 19
GND 22
2.2K
56K
3.3K
.1
5.6K
+15V
+15V
6 5 7
2.2K
2.2K
2.2K
270
1.2K
47µ
10K
10µ
10K
.005
130
10K
50µ
1.5K
50K
10K
1K
To. 8
A
CA-3019
330
30µ
5µh
50µ
B
270
5µ
22K
22K
75µ
1K
75
12K
33K
68K
To. 16
50
75
1.2K
1.5K
5.6K
100µ
6V
22K
10K
820
2.2K
100µ
50µ
100µ
56K
1.2K
5.6K
120
620
100µ
A
620
100µ
B
3.3K
PED. CONT.


1
2
3
IN 4
5
OUT 6
7
8
IN 9
10
OUT 11
12
13
14
15
IN 16
17
OUT 18
19
20
21
22
5μh
-6V
100μ
5μh
+12V
100μ


+12V
1350
10K
51
+6V
+100μ
OUT
75
IN
S1-D
1350
1K
-6V
-6V

Mixing Amp


1
2
-6V
VID IN 3
+100μ
4
5
OUT 6
7
8
9
IN 10
11
12
OUT 13
14
15
16
IN 17
18
19
OUT 20
21
+12V
22
+100μ


+12V
IN
100
10M
CA-2023
500μ
75
OUT
820
1K
75μ
1.2K
75
2K
-6V

Amp.